

Maximum Likelihood Estimation Methods

Maximum likelihood estimation (MLE) methods are now among the most popular, best understood and most efficient estimation methods for long-range dependent time series. In this chapter, we provide a basic overview of the methods, as these are indispensable when working with long-range dependence, and improve on the heuristic methods discussed in Section 2.10. A more thorough treatment can be found in statistically oriented works (Palma [789], Giraitis et al.[406], Beran et al.[127]). We consider:

- The exact MLE in the time domain (Section 10.1)
- Whittle estimation in the spectral domain (Section 10.2.1)
- Autoregressive approximation (Section 10.2.2)
- The local Whittle estimation (Section 10.6.1)
- The broadband Whittle approach (Section 10.7)

10.1 Exact Gaussian MLE in the Time Domain

Suppose $\{X_n\}_{n \in \mathbb{Z}}$ is a Gaussian, zero mean, stationary series with autocovariance function $\gamma_\eta(n)$ and spectral density $f_\eta(\lambda)$. Here, η denotes a parameter vector, supposed to be unknown, which characterizes the series $\{X_n\}_{n \in \mathbb{Z}}$. (Unlike in the previous chapter, for notational simplicity, we shall suppress the subscript X in γ and f , by writing γ_η and f_η instead of $\gamma_{X,\eta}$ and $f_{X,\eta}$.) The vector η_0 will denote the *true value* of the parameter vector.

Example 10.1.1 (*Parametric FARIMA(p, d, q) family*) A Gaussian, zero mean, stationary FARIMA(p, d, q), $d \in (-1/2, 1/2)$, series is characterized by the spectral density

$$f_\eta(\lambda) = \frac{\sigma_Z^2}{2\pi} |1 - e^{-i\lambda}|^{-2d} \frac{|\theta(e^{-i\lambda})|^2}{|\varphi(e^{-i\lambda})|^2}, \quad \lambda \in (-\pi, \pi], \quad (10.1.1)$$

where

$$\varphi(z) = 1 - \varphi_1 z - \cdots - \varphi_p z^p, \quad \theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$$

and hence

$$\eta = (\sigma_Z^2, d, \varphi_1, \dots, \varphi_p, \theta_1, \dots, \theta_q)' \in \mathbb{R}^{p+q+2},$$

where prime indicates transpose (see Section 2.4.2). The autocovariance function $\gamma_\eta(n)$ does not have an explicit form in general but could be computed numerically in an efficient way (Sowell [918], Chung [242], Bertelli and Caporin [136], Doornik and Ooms [326]). See also Exercise 10.1.

Given N observations $X_1, \dots, X_N$ of the process $\{X_n\}_{n \in \mathbb{Z}}$, form the vector $X = (X_1, \dots, X_N)'$. The (Gaussian) *likelihood function* is defined as

$$L(\eta) = (2\pi)^{-\frac{N}{2}} \det(\Sigma_N(\eta))^{-\frac{1}{2}} \exp \left\{ -\frac{1}{2} X' \Sigma_N(\eta)^{-1} X \right\},$$

where $\Sigma_N(\eta) = (\gamma_\eta(j-k))_{j,k=1,\dots,N}$. The *log-likelihood* is

$$\ell(\eta) = -\frac{N}{2} \log 2\pi - \frac{1}{2} \log \det(\Sigma_N(\eta)) - \frac{1}{2} X' \Sigma_N(\eta)^{-1} X.$$

After multiplying by $-\frac{2}{N}$ and ignoring the constant term, the *maximum likelihood estimator* of η is defined as

$$\hat{\eta}_{mle} = \underset{\eta}{\operatorname{argmin}} \ell_{exact}(\eta), \quad (10.1.2)$$

where the (*scaled*) *negative log-likelihood* is

$$\ell_{exact}(\eta) = \frac{1}{N} \log \det(\Sigma_N(\eta)) + \frac{1}{N} X' \Sigma_N(\eta)^{-1} X. \quad (10.1.3)$$

Remark 10.1.2 The series is assumed to have zero mean μ for simplicity. The unknown mean μ could be incorporated in the MLE as another parameter by replacing X by $X - \mu$. The resulting ML estimator of μ will be a linear combination of the components of the vector X . It is common, however, to estimate the mean μ by the sample mean $\bar{X}$ which is itself a linear combination of the components of X . When the components of X are dependent, it is likewise natural to consider the Best Linear Unbiased Estimator (BLUE) of the mean. It is known (Samarov and Taquq [877]) that for LRD series, the BLUE provides only minimal gains in efficiency when compared to the sample mean (see also Exercise 10.2). This further justifies the simplification $\mu = 0$, we make for the MLE. It should also be noted that some asymptotic results for ML estimators have been derived when X is replaced by $X - \bar{X}$ or $X - \hat{\mu}$ with a mean estimator $\hat{\mu}$ satisfying suitable conditions (Dahlhaus [277, 278], Lieberman, Rosemarin, and Rousseau [631]). For further remarks, see also Remark 10.2.5. Finally, we also point out that in Section 10.2.1, the MLE is approximated in the frequency domain. In the latter domain, the approximate MLE is invariant to an additive shift of the data and hence $\mu = 0$ can be assumed.

The MLE $\hat{\eta}_{mle}$ is computed through a numerical optimization. We are not going to discuss optimization methods here (see, for example, Lange [592]). But any optimization procedure involves computation of the (negative) log-likelihood $\ell_{exact}(\eta)$ for a given value η , which in turn requires computing $\Sigma_N(\eta)$, its inverse $\Sigma_N(\eta)^{-1}$ and the determinant $\det(\Sigma_N(\eta))$. Computing $\Sigma_N(\eta)$ may not be trivial itself (see Example 10.1.1). But it is the computation of $\Sigma_N(\eta)^{-1}$ and $\det(\Sigma_N(\eta))$ which presents real challenges, especially for larger sample sizes N .

Naïve ways to compute $\Sigma_N(\eta)^{-1}$, for example, through Cholesky decomposition, require $O(N^3)$ number of steps and are impractical already for moderate sample sizes N ($N \approx 500$). The computations could be sped up to the $O(N^2)$ number of steps by exploiting the Toeplitz structure of the covariance matrix $\Sigma_N(\eta)$, which allows one to deal with longer time series (up to $N \approx 10,000$). Two such popular methods, which are important for other

considerations as well, are the Durbin-Levinson and innovations algorithms (e.g., Brockwell and Davis [186]).

- The *Durbin-Levinson algorithm* provides a recursive way (in n below), given an autocovariance function, to compute the coefficients $\varphi_{n1}, \dots, \varphi_{nn}$ in

$$\widehat{X}_{n+1} = \varphi_{n1}X_n + \dots + \varphi_{nn}X_1, \quad n = 1, 2, \dots, \quad (10.1.4)$$

where $\widehat{X}_{n+1}$ is the best linear one-step predictor of X_{n+1} in $L^2(\Omega)$ based on $X_1, \dots, X_n$, and also obtain the mean squared error of the prediction

$$v_n = \mathbb{E}(X_{n+1} - \widehat{X}_{n+1})^2, \quad n = 1, 2, \dots \quad (10.1.5)$$

One sets $\widehat{X}_1 = 0$ and $v_0 = \mathbb{E}X_1^2$ by convention.

- The *innovations algorithm* provides a recursive way (in n), given an autocovariance function, to compute the coefficients $\theta_{n1}, \dots, \theta_{nn}$ in

$$\widehat{X}_{n+1} = \theta_{n1}(X_n - \widehat{X}_n) + \dots + \theta_{nn}(X_1 - \widehat{X}_1), \quad n = 1, 2, \dots, \quad (10.1.6)$$

and the v_n s. Note that (10.1.6) can be written as

$$X_{n+1} = \theta_{nn}(X_1 - \widehat{X}_1) + \dots + \theta_{n1}(X_n - \widehat{X}_n) + (X_{n+1} - \widehat{X}_{n+1}), \quad n = 1, 2, \dots \quad (10.1.7)$$

The components $X_i - \widehat{X}_i$ and $X_j - \widehat{X}_j$ are uncorrelated for $i \neq j$, and hence

$$\mathbb{E}(X - \widehat{X})(X - \widehat{X})' = \text{diag}(v_0, \dots, v_{N-1}) =: D_N, \quad (10.1.8)$$

where $\widehat{X} = (\widehat{X}_1, \dots, \widehat{X}_N)'$.

For a detailed description of the Durbin-Levinson and innovations algorithms, see Brockwell and Davis [186], Chapter 5.

To see how these algorithms are used to get ℓ_{exact} , note that (10.1.7) can be expressed in a matrix form as

$$X = C_N(X - \widehat{X}), \quad (10.1.9)$$

where

$$C_N = \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 \\ \theta_{11} & 1 & 0 & 0 & \dots & 0 & 0 \\ \theta_{22} & \theta_{21} & 1 & 0 & \dots & 0 & 0 \\ \theta_{33} & \theta_{32} & \theta_{31} & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \theta_{N-1,N-1} & \theta_{N-1,N-2} & \theta_{N-1,N-3} & \theta_{N-1,N-4} & \dots & \theta_{N-1,1} & 1 \end{pmatrix}$$

and $\widehat{X} = (\widehat{X}_1, \dots, \widehat{X}_N)'$ as above. By using (10.1.8), one has

$$\Sigma_N(\eta) = \mathbb{E}XX' = C_N\mathbb{E}(X - \widehat{X})(X - \widehat{X})'C_N' = C_N D_N C_N'.$$

Then,

$$\begin{aligned} X' \Sigma_N(\eta)^{-1} X &= (X - \widehat{X})' C_N' (C_N D_N C_N')^{-1} C_N (X - \widehat{X}) \\ &= (X - \widehat{X})' D_N^{-1} (X - \widehat{X}) = \sum_{j=1}^N \frac{(X_j - \widehat{X}_j)^2}{v_{j-1}} \end{aligned}$$

and

$$\det(\Sigma_N(\eta)) = \det(C_N) \det(D_N) \det(C_N') = \det(D_N) = \prod_{j=1}^N v_{j-1},$$

since the determinant of a triangular matrix equals the product of the elements in the diagonal. Hence, the log-likelihood (10.1.3) can be written as

$$\ell_{exact}(\eta) = \frac{1}{N} \sum_{j=1}^N \log v_{j-1} + \frac{1}{N} \sum_{j=1}^N \frac{(X_j - \widehat{X}_j)^2}{v_{j-1}}, \quad (10.1.10)$$

where the $X_j - \widehat{X}_j$ s and v_j s can be computed, for a fixed η and the resulting autocovariance function γ_η , by using either the Durbin-Levinson or the innovations algorithm (see also the discussion at the end of Example 10.1.1 for FARIMA models).

Another difficulty with the exact MLE often raised in the literature, even for small sample sizes, is that $\Sigma_N(\eta)$ can become ill-conditioned when the LRD parameter d is close to $1/2$, making the computation of the inverse numerically unstable. The degree of ill-conditioning can be measured by the ratio of the largest and smallest eigenvalues of the matrix, called the condition number (e.g., Gentle [385], Monahan [728]). A large condition number indicates that a matrix is ill-conditioned. For example, for FARIMA(0, d , 0) series and $N = 100$, by using the explicit formula for the autocovariance function of FARIMA(0, d , 0) series in Corollary 2.4.4, it can be checked numerically that the condition number is 8.35, 28.71, 133.96, 411.98 for $d = 0.2, 0.3, 0.4, 0.45$, respectively.

The difficulties with the exact MLE discussed above (computational issues for larger sample sizes and when LRD parameter approaches $1/2$) call for approximations of the likelihood. Two such approximations are discussed in the next section. Asymptotic results for the exact MLE will be discussed briefly in the context of those approximations.

10.2 Approximate MLE

Two approximations of the log-likelihood (10.1.3) are discussed below: the celebrated Whittle's approximation in the spectral domain (Section 10.2.1) and that based on the autoregressive representation (Section 10.2.2).

10.2.1 Whittle Estimation in the Spectral Domain

The Whittle's approximation deals with the two terms, $\frac{1}{N} \log \det(\Sigma_N(\eta))$ and $\frac{1}{N} X' \Sigma_N(\eta)^{-1} X$, of the log-likelihood (10.1.3) separately.

Approximation of $\frac{1}{N} \log \det(\Sigma_N(\eta))$: This term is approximated by using Szegő's limit theorem; that is,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log \det(\Sigma_N(\eta)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \log(2\pi f_{\eta}(\lambda)) d\lambda \quad (10.2.1)$$

(cf. the strong Szegő limit theorem in (3.11.101)). The presence of 2π next to $f_{\eta}(\lambda)$ is to account for the fact that the Fourier transform of $\{\gamma_{\eta}(n)\}$ (see Appendix A.1.1) is $2\pi f_{\eta}(\lambda)$, and not $f_{\eta}(\lambda)$.

Approximation of $\frac{1}{N} X^T \Sigma_N(\eta)^{-1} X$: Set $A_N(\eta) = (\alpha_{\eta}(j-k))_{j,k=1,\dots,N}$, where

$$\alpha_{\eta}(j-k) = \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} e^{i(j-k)\lambda} \frac{1}{f_{\eta}(\lambda)} d\lambda,$$

that is, $\hat{\alpha}_{\eta}(\lambda) = f_{\eta}(\lambda)^{-1}/(2\pi)$, where the hat indicates the Fourier transform (see Appendix A.1.1). Then, for large N , $A_N(\eta)$ approximates $\Sigma_N(\eta)^{-1}$. Indeed, note that $\Sigma_N(\eta) = (\gamma_{\eta}(j-k))_{j,k=1,\dots,N}$, where

$$\gamma_{\eta}(j-k) = \int_{-\pi}^{\pi} e^{i(j-k)\lambda} f_{\eta}(\lambda) d\lambda,$$

that is, $\hat{\gamma}_{\eta}(\lambda) = 2\pi f_{\eta}(\lambda)$. Then, for large N , we expect that

$$\begin{aligned} (A_N(\eta) \Sigma_N(\eta))_{j,k} &= \sum_{l=1}^N \alpha_{\eta}(j-l) \gamma_{\eta}(l-k) \approx \sum_{l=-\infty}^{\infty} \alpha_{\eta}(j-l) \gamma_{\eta}(l-k) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i(j-k)\lambda} \hat{\alpha}_{\eta}(\lambda) \overline{\hat{\gamma}_{\eta}(\lambda)} d\lambda = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i(j-k)\lambda} \frac{1}{f_{\eta}(\lambda)} f_{\eta}(\lambda) d\lambda = \begin{cases} 1, & \text{if } j=k, \\ 0, & \text{if } j \neq k, \end{cases} \end{aligned}$$

by using Parseval's identity (A.1.3) and the relations between $\hat{\alpha}_{\eta}(\lambda)$, $\hat{\gamma}_{\eta}(\lambda)$ and $f_{\eta}(\lambda)$ noted above, where $\approx$ indicates an approximation for large N . This leads to the approximation

$$\begin{aligned} \frac{1}{N} X^T \Sigma_N(\eta)^{-1} X &\approx \frac{1}{N} X^T A_N(\eta) X = \frac{1}{N} \sum_{j,k=1}^N \alpha_{\eta}(j-k) X_j X_k \\ &= \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \frac{\frac{1}{N} |\sum_{j=1}^N e^{ij\lambda} X_j|^2}{f_{\eta}(\lambda)} d\lambda = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{I_X(\lambda)}{2\pi f_{\eta}(\lambda)} d\lambda, \quad (10.2.2) \end{aligned}$$

where $I_X(\lambda)$ is the periodogram (2.10.9) (see also the warning in Section 1.3.3).

Combining (10.2.1) and (10.2.2), consider the Whittle approximation of the log-likelihood (10.1.3)

$$\ell_{\text{whittle}}(\eta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \log(2\pi f_{\eta}(\lambda)) d\lambda + \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{I_X(\lambda)}{2\pi f_{\eta}(\lambda)} d\lambda \quad (10.2.3)$$

and define

$$\hat{\eta}_{\text{whittle}} = \underset{\eta}{\operatorname{argmin}} \ell_{\text{whittle}}(\eta). \quad (10.2.4)$$

The discrete version of $\ell_{whittle}(\eta)$ is

$$\sum_{j=1}^N \log \left(2\pi f_{\eta} \left(\frac{2\pi j}{N} \right) \right) + \sum_{j=1}^N \frac{I_X \left(\frac{2\pi j}{N} \right)}{f_{\eta} \left(\frac{2\pi j}{N} \right)}, \quad (10.2.5)$$

which can be computed using the Fast Fourier Transform (FFT) in $O(N \log N)$ steps. (The 2π in the first integral in (10.2.3) and the first sum in (10.2.5) does not affect the optimization but we keep them for cancellation below.)

An important *special case* of a parametric family of spectral densities is

$$f_{\eta}(\lambda) = \frac{\sigma^2}{2\pi} |g_v(e^{-i\lambda})|^2 = \frac{\sigma^2}{2\pi} \left| \sum_{j=0}^{\infty} g_{v,j} e^{-ij\lambda} \right|^2 =: \frac{\sigma^2}{2\pi} h_v(\lambda), \quad (10.2.6)$$

where $g_{v,0} = 1$, $\eta = (\sigma^2, v)'$ and $v \in \mathbb{R}^q$. Note that the parameter σ^2 appears as a multiplicative factor in (10.2.6). This special case (10.2.6) corresponds to a time series with a linear representation

$$X_n = \sum_{j=0}^{\infty} g_{v,j} Z_{n-j},$$

where $\{Z_n\}$ is a white noise series with variance $\sigma^2 = \sigma_Z^2$. For example, the parametric family of FARIMA(p, d, q) spectral densities (10.1.1) has the form (10.2.6) with $v = (d, \varphi_1, \dots, \varphi_p, \theta_1, \dots, \theta_q)'$. Note that g_v is a function of $e^{-i\lambda}$ while h_v is a function of λ .

In the case (10.2.6), one has

$$\int_{-\pi}^{\pi} \log h_v(\lambda) d\lambda = 0. \quad (10.2.7)$$

This can be seen, on one hand, as a consequence of Kolmogorov's formula (e.g., Brockwell and Davis [186], Section 5.8), which asserts that $\int_{-\pi}^{\pi} \log f_{\eta}(\lambda) d\lambda = 2\pi \log(\sigma^2/2\pi)$. By (10.2.6), $\log f_{\eta} = \log(\sigma^2/2\pi) + \log h_v$ and therefore $\int_{-\pi}^{\pi} \log h_v(\lambda) d\lambda = \int_{-\pi}^{\pi} \log f_{\eta}(\lambda) d\lambda - \int_{-\pi}^{\pi} \log(\sigma^2/2\pi) d\lambda = 2\pi \log(\sigma^2/2\pi) - 2\pi \log(\sigma^2/2\pi) = 0$. On the other hand, this could informally be explained as follows. Note that $\log h_v(\lambda) = \log g_v(e^{-i\lambda}) + \log g_v(e^{i\lambda})$. Consider the expansion $g_v(z) = \sum_{j=0}^{\infty} g_{v,j} z^j$ in (10.2.6). Since $g_v(0) = g_{v,0} = 1$, one has $\log g_v(0) = 0$ and hence the expansion $\log g_v(z) = \sum_{k=1}^{\infty} a_{v,k} z^k$ starts at $k = 1$. We now obtain that, for example, $\log g_v(e^{-i\lambda}) = \sum_{k=1}^{\infty} a_{v,k} e^{-ik\lambda}$ and hence that $\int_{-\pi}^{\pi} \log g_v(e^{-i\lambda}) d\lambda = \sum_{k=1}^{\infty} a_{v,k} \int_{-\pi}^{\pi} e^{-ik\lambda} d\lambda = 0$.

By using (10.2.7), the log-likelihood (10.2.3) can be written as

$$\ell_{whittle}(\sigma^2, v) = \log \sigma^2 + \frac{1}{\sigma^2} \int_{-\pi}^{\pi} \frac{I_X(\lambda)}{2\pi h_v(\lambda)} d\lambda \quad (10.2.8)$$

and the corresponding estimator of (σ^2, v) as

$$(\hat{\sigma}_{whittle}^2, \hat{v}_{whittle}) = \underset{\sigma^2, v}{\operatorname{argmin}} \ell_{whittle}(\sigma^2, v). \quad (10.2.9)$$

The minimization could also be carried over v alone after solving for σ^2 (in terms of v) and substituting it into $\ell_{whittle}(\sigma^2, v)$.

Another perspective on the log-likelihood in (10.2.8) is given in Exercise 10.3.

In the following result, we present the large sample asymptotics of the Whittle estimators (10.2.9) in the case (10.2.6). The following assumptions will be used on the functions h_v defined in (10.2.6):

(A0) The parameter space Υ of v is compact, the parameters $(\sigma, v) \in (0, \infty) \times \Upsilon$ determine the spectral density $\frac{\sigma^2}{2\pi} h_v$ uniquely, and for any $a > 0$, the function $1/(h_v(\lambda) + a)$ is continuous in $(\lambda, v) \in (-\pi, \pi) \times \Upsilon$. Moreover, the true value (σ_0, v_0) lies in the interior of $(0, \infty) \times \Upsilon$.

There is, moreover, a compact ball Υ_0 centered at v_0 such that:

(A1) $\int_{-\pi}^{\pi} \log h_v(\lambda) d\lambda (\equiv 0)$ is twice differentiable in $v \in \Upsilon_0$ under the integral sign.

(A2) The functions $h_v(\lambda)$, $\frac{\partial}{\partial v_j} h_v^{-1}(\lambda)$ and $\frac{\partial^2}{\partial v_r \partial v_s} h_v^{-1}(\lambda)$ are continuous at all $\lambda \in (-\pi, \pi) \setminus \{0\}$, $v \in \Upsilon_0$, for $r, s = 1, \dots, q$.

(A3) There are $-1 < \alpha, \beta < 1$ such that $\alpha + \beta < 1/2$ and

$$h_{v_0}(\lambda) \leq C|\lambda|^{-\alpha}, \quad \left| \frac{\partial}{\partial \lambda} h_{v_0}(\lambda) \right| \leq C|\lambda|^{-\alpha-1}, \quad \left| \frac{\partial}{\partial v_j} h_{v_0}^{-1}(\lambda) \right| \leq C|\lambda|^{-\beta},$$

for $j = 1, \dots, q$, $\lambda \in (-\pi, \pi) \setminus \{0\}$.

(A4) If $\alpha \leq 0$, then there is $G \in L^1(-\pi, \pi)$ such that $\left| \frac{\partial^2}{\partial v_r \partial v_s} h_v^{-1}(\lambda) \right| \leq G(\lambda)$, for $\lambda \in (-\pi, \pi) \setminus \{0\}$, $v \in \Upsilon_0$, $r, s = 1, \dots, q$. If $\alpha > 0$, then $\frac{\partial^2}{\partial v_r \partial v_s} h_v^{-1}(\lambda)$ is continuous in $\lambda \in (-\pi, \pi) \setminus \{0\}$, $v \in \Upsilon_0$, for $r, s = 1, \dots, q$.

Theorem 10.2.1 Suppose that $\{X_n\}_{n \in \mathbb{Z}}$ has a linear representation

$$X_n = \sum_{k=0}^{\infty} g_{v_0, k} \epsilon_{n-k},$$

where $\{\epsilon_n\}_{n \in \mathbb{Z}}$ consists of i.i.d. variables with zero mean, variance σ_0^2 and finite fourth moment. Suppose its spectral density belongs to the family given by (10.2.6) where the functions h_v satisfy Assumptions (A0)–(A4) above. Then, as $N \rightarrow \infty$,

$$\sqrt{N}(\hat{v}_{\text{whittle}} - v_0) \xrightarrow{d} \mathcal{N}(0, 4\pi C^{-1}(v_0)), \quad (10.2.10)$$

$$\sqrt{N}(\hat{\sigma}_{\text{whittle}}^2 - \sigma_0^2) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}\epsilon_0^4 - \sigma_0^4), \quad (10.2.11)$$

where $C(v) = (c_{rs}(v))_{r,s=1,\dots,q}$ with

$$c_{rs}(v) = \int_{-\pi}^{\pi} \frac{\partial}{\partial v_r} \log h_v(\lambda) \frac{\partial}{\partial v_s} \log h_v(\lambda) d\lambda. \quad (10.2.12)$$

Remark 10.2.2 The assumptions (A0)–(A4) of Theorem 10.2.1 allow for LRD time series such as FARIMA(p, d, q) series in Example 10.1.1 (Proposition 8.3.1, Giraitis et al. [406]). The theorem shows that the rate $\sqrt{N}$ holds even for the LRD parameter estimator $\hat{d}_{\text{whittle}}$. Note that the series $\{X_n\}$ is not necessarily Gaussian. It can also be shown that $(\hat{\sigma}_{\text{whittle}}^2, \hat{v}_{\text{whittle}})$ are jointly asymptotically normal with zero asymptotic covariances (Beran et al. [127], p. 425).

Remark 10.2.3 Under suitable assumptions, the asymptotic results (10.2.10)–(10.2.11) also hold for the exact Gaussian MLE estimators $(\hat{\sigma}_{mle}^2, \hat{\nu}_{mle}) = \hat{\eta}_{mle}$ (Dahlhaus [277, 278], Lieberman et al. [631]), and for the Whittle estimators when using the discrete version of the log-likelihood (10.2.5) (see the discussion following Remark 2.2 in Dahlhaus [277], and Velasco and Robinson [988]).

Proof outline of Theorem 10.2.1 We provide here an outline of the proof of the theorem. A complete proof can be found in Giraitis et al. [406], Theorem 3.8.1. The proof is standard in the context of MLE though technical issues arise related to the convergence of functionals involving LRD series.

The MLE estimators $(\hat{\sigma}_{whittle}^2, \hat{\nu}_{whittle})$ arise from minimizing $\ell_{whittle}(\sigma^2, \nu)$ in (10.2.8). Note that, in view of (2.10.9), we have

$$\ell_{whittle}(\sigma^2, \nu) = \log \sigma^2 + \frac{1}{\sigma^2 2\pi N} \sum_{j,k=1}^N \beta_\nu(j-k) X_j X_k, \quad (10.2.13)$$

where

$$\beta_\nu(n) = \int_{-\pi}^{\pi} e^{in\lambda} \frac{1}{h_\nu(\lambda)} d\lambda. \quad (10.2.14)$$

We first consider the asymptotics of $\hat{\nu}_{whittle}$. One can show that, under Assumption (A1), $\hat{\nu}_{whittle}$ is consistent (Giraitis et al. [406], Theorem 8.2.1). Then, for large enough N and with probability approaching 1, $\hat{\nu}_{whittle} \in \Upsilon_0$ and satisfies the first order condition

$$\dot{Q}(\hat{\nu}_{whittle}) = 0,$$

where

$$\dot{Q}(\nu) = \frac{1}{N} \sum_{j,k=1}^N \dot{\beta}_\nu(j-k) X_j X_k \quad (10.2.15)$$

and $\dot{\beta}_\nu = \partial \beta_\nu / \partial \nu$. (Note that, while $Q(\nu)$ is a scalar, $\dot{Q}(\nu)$ has the dimension of the vector ν .) By Taylor's expansion,

$$0 = \dot{Q}(\hat{\nu}_{whittle}) = \dot{Q}(\nu_0) + \ddot{Q}(\nu^*)(\hat{\nu}_{whittle} - \nu_0),$$

where $\ddot{Q} = \partial \dot{Q} / \partial \nu$ is a matrix and $|\nu^* - \nu_0| \leq |\hat{\nu}_{whittle} - \nu_0|$. Hence,

$$\sqrt{N}(\hat{\nu}_{whittle} - \nu_0) = -\sqrt{N}(\ddot{Q}(\nu^*))^{-1} \dot{Q}(\nu_0). \quad (10.2.16)$$

We next consider the convergence of $\ddot{Q}(\nu^*)$ and $\sqrt{N} \dot{Q}(\nu_0)$. The matrix $\ddot{Q}(\nu^*)$ consists of elements

$$\ddot{Q}(\nu^*)_{rs} = \frac{1}{N} \sum_{j,k=1}^N \frac{\partial^2}{\partial \nu_r \partial \nu_s} \beta_\nu(j-k) X_j X_k.$$

In view of (10.2.14) and (2.10.9), we have

$$\ddot{Q}(\nu^*)_{rs} = \int_{-\pi}^{\pi} I_X(\lambda) \frac{\partial^2}{\partial \nu_r \partial \nu_s} \left(\frac{1}{h_\nu(\lambda)} \right) \Big|_{\nu=\nu^*} d\lambda.$$

Since v^* converges to v_0 in probability and $I_X(\lambda)$ is an estimator of $2\pi f_{(\sigma_0^2, v_0)}(\lambda)$, it is expected and can be shown that

$$\begin{aligned}\ddot{Q}(v^*)_{rs} &\xrightarrow{P} 2\pi \int_{-\pi}^{\pi} f_{(\sigma_0^2, v_0)}(\lambda) \frac{\partial^2}{\partial v_r \partial v_s} \left(\frac{1}{h_v(\lambda)} \right) \Big|_{v=v_0} d\lambda = \sigma_0^2 \int_{-\pi}^{\pi} h_v(\lambda) \frac{\partial^2}{\partial v_r \partial v_s} \left(\frac{1}{h_v(\lambda)} \right) \Big|_{v=v_0} d\lambda \\ &= \sigma_0^2 \int_{-\pi}^{\pi} \left(\frac{2}{h_v(\lambda)^2} \frac{\partial}{\partial v_r} h_v(\lambda) \frac{\partial}{\partial v_s} h_v(\lambda) - \frac{1}{h_v(\lambda)} \frac{\partial^2}{\partial v_r \partial v_s} h_v(\lambda) \right) \Big|_{v=v_0} d\lambda.\end{aligned}\quad (10.2.17)$$

Note that, by differentiating (10.2.7) twice inside the integral sign, we also have

$$0 = \int_{-\pi}^{\pi} \left(\frac{1}{h_v(\lambda)^2} \frac{\partial}{\partial v_r} h_v(\lambda) \frac{\partial}{\partial v_s} h_v(\lambda) - \frac{1}{h_v(\lambda)} \frac{\partial^2}{\partial v_r \partial v_s} h_v(\lambda) \right) d\lambda. \quad (10.2.18)$$

Combining (10.2.17) and (10.2.18) yields

$$\begin{aligned}\ddot{Q}(v^*)_{rs} &\xrightarrow{P} \sigma_0^2 \int_{-\pi}^{\pi} \frac{1}{h_v(\lambda)^2} \frac{\partial}{\partial v_r} h_v(\lambda) \frac{\partial}{\partial v_s} h_v(\lambda) \Big|_{v=v_0} d\lambda \\ &= \sigma_0^2 \int_{-\pi}^{\pi} \frac{\partial}{\partial v_r} \log h_v(\lambda) \frac{\partial}{\partial v_s} \log h_v(\lambda) \Big|_{v=v_0} d\lambda = \sigma_0^2 c_{rs}(v_0),\end{aligned}\quad (10.2.19)$$

where $c_{rs}(v)$ appears in (10.2.12).

The term $\sqrt{N} \dot{Q}(v_0)$, on the other hand, can be shown to be asymptotically normal. In view of (10.2.15), $\sqrt{N} \dot{Q}(v_0)$ is referred to as a quadratic form. The convergence of quadratic forms to a normal limit has attracted considerable attention in the literature on long-range dependent series. For $\sqrt{N} \dot{Q}(v_0)$, it can be shown that

$$\sqrt{N}(\dot{Q}(v_0) - \mathbb{E} \dot{Q}(v_0)) \xrightarrow{d} \mathcal{N}(0, 4\pi \sigma_0^4 C(v_0)) \quad (10.2.20)$$

and that

$$\sqrt{N} \mathbb{E} \dot{Q}(v_0) \rightarrow 0. \quad (10.2.21)$$

By using (10.2.20), (10.2.21) and (10.2.19), the relation (10.2.16) implies the desired convergence (10.2.10).

The asymptotic normality (10.2.20) is proved by showing first (Giraitis et al. [406], Lemma 6.3.1) that

$$N \text{Var}(\dot{Q}(v_0) - R(v_0)) \rightarrow 0, \quad (10.2.22)$$

where

$$R(v) = \frac{1}{N} \sum_{k=1}^N Y_{1,k} Y_{2,k}, \quad Y_{j,k} = \sum_{m=-\infty}^{\infty} \ell_{j,v,m} \epsilon_{k-m}, \quad j = 1, 2, \quad (10.2.23)$$

and where the Fourier transforms of $\{\ell_{j,v_0,m}\}$ with m as variable, are

$$\begin{aligned}\widehat{\ell}_{1,v}(\lambda) &= (2\pi)^{1/2} g_v(e^{-i\lambda}) \left| \frac{\partial}{\partial v} \frac{1}{h_v(\lambda)} \right|^{\frac{1}{2}}, \\ \widehat{\ell}_{2,v}(\lambda) &= (2\pi)^{1/2} g_v(e^{-i\lambda}) \left| \frac{\partial}{\partial v} \frac{1}{h_v(\lambda)} \right|^{\frac{1}{2}} \text{sign} \left(\frac{\partial}{\partial v} \frac{1}{h_v(\lambda)} \right)\end{aligned}\quad (10.2.24)$$

(the square root above, as well as the multiplications of $Y_{1,k} Y_{2,k}$ and by the sign function, are taken componentwise). Recall that g_v and h_v are related through (10.2.6). It is

then enough to prove (10.2.20) with $\dot{Q}(\nu_0)$ replaced by $R(\nu_0)$. Note that, in contrast to the series X_k entering $\dot{Q}(\nu)$ which may be long-range dependent, the series $Y_{j,k}$ are expected to have shorter memory. For example, with $g_\nu(e^{-i\lambda}) = (1 - e^{-i\lambda})^{-d}$ corresponding to FARIMA(0, d , 0) series and $\nu = d$, we have $|\widehat{\ell}_{1,\nu}(\lambda)|^2 = |\log|1 - e^{-i\lambda}||$ which is now bounded by $C|\lambda|^{-\delta}$ around $\lambda = 0$ for any arbitrarily small $\delta > 0$ (see Example 10.2.4 below).

We shall not prove the asymptotic normality (10.2.20) with $\dot{Q}(\nu_0)$ replaced by $R(\nu_0)$. We shall just argue for the asymptotic covariance matrix $4\pi\sigma_0^4 C(\nu_0)$. By using the superscript (r) to denote the r th element of a vector, note that

$$\begin{aligned} & \text{Cov}\left(\frac{1}{N^{1/2}} \sum_{k=1}^N Y_{1,k}^{(r)} Y_{2,k}^{(r)}, \frac{1}{N^{1/2}} \sum_{k=1}^N Y_{1,k}^{(s)} Y_{2,k}^{(s)}\right) \\ &= \frac{1}{N} \sum_{k=-(N-1)}^{N-1} (N - |k|) \text{Cov}\left(Y_{1,0}^{(r)} Y_{2,0}^{(r)}, Y_{1,k}^{(s)} Y_{2,k}^{(s)}\right) \rightarrow \sum_{k=-\infty}^{\infty} \text{Cov}\left(Y_{1,0}^{(r)} Y_{2,0}^{(r)}, Y_{1,k}^{(s)} Y_{2,k}^{(s)}\right) =: W_{rs}, \end{aligned}$$

where the first equality is obtained analogously as in the proof of Proposition 2.2.5. We show next that the asymptotic covariance W_{rs} is indeed $4\pi\sigma_0^4 c_{rs}(\nu_0)$.

Writing $Y_{1,k}^{(r)} Y_{2,k}^{(r)} = \sum_{m_1, m_2=-\infty}^{\infty} \ell_{1,\nu_0,k-m_1}^{(r)} \ell_{2,\nu_0,k-m_2}^{(r)} \epsilon_{m_1} \epsilon_{m_2}$, we have

$$W_{rs} = \sum_{k=-\infty}^{\infty} \sum_{m_1, m_2=-\infty}^{\infty} \sum_{n_1, n_2=-\infty}^{\infty} \ell_{1,\nu_0,-m_1}^{(r)} \ell_{2,\nu_0,-m_2}^{(r)} \ell_{1,\nu_0,k-n_1}^{(s)} \ell_{2,\nu_0,k-n_2}^{(s)} \text{Cov}(\epsilon_{m_1} \epsilon_{m_2}, \epsilon_{n_1} \epsilon_{n_2}).$$

Note that $\text{Cov}(\epsilon_{m_1} \epsilon_{m_2}, \epsilon_{n_1} \epsilon_{n_2}) \neq 0$ only in the following three cases:

(i) $m_1 = m_2 = n_1 = n_2$, (ii) $m_1 = n_1 \neq m_2 = n_2$, and (iii) $m_1 = n_2 \neq m_2 = n_1$.

Denote the contribution of these cases to W_{rs} by $W_{1,rs}$, $W_{2,rs}$ and $W_{3,rs}$, respectively.

Since $\text{Cov}(\epsilon_0 \epsilon_0, \epsilon_0 \epsilon_0) = \text{Var}(\epsilon_0^2)$, then setting $n = m_1 = m_2 = n_1 = n_2$, we have

$$\begin{aligned} W_{1,rs} &= \text{Var}(\epsilon_0^2) \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \ell_{1,\nu_0,-n}^{(r)} \ell_{2,\nu_0,-n}^{(r)} \ell_{1,\nu_0,k-n}^{(s)} \ell_{2,\nu_0,k-n}^{(s)} \\ &= \text{Var}(\epsilon_0^2) \sum_{n=-\infty}^{\infty} \ell_{1,\nu_0,n}^{(r)} \ell_{2,\nu_0,n}^{(r)} \sum_{k=-\infty}^{\infty} \ell_{1,\nu_0,k}^{(s)} \ell_{2,\nu_0,k}^{(s)}. \end{aligned}$$

By using Parseval's identity (A.1.3), (10.2.24), (10.2.6) and (10.2.7), we conclude that

$$\begin{aligned} W_{1,rs} &= \text{Var}(\epsilon_0^2) \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{\ell}_{1,\nu_0}^{(r)}(\lambda) \overline{\widehat{\ell}_{2,\nu_0}^{(r)}(\lambda)} d\lambda \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{\ell}_{1,\nu_0}^{(s)}(\lambda) \overline{\widehat{\ell}_{2,\nu_0}^{(s)}(\lambda)} d\lambda \\ &= \text{Var}(\epsilon_0^2) \int_{-\pi}^{\pi} \left(|g_\nu(e^{-i\lambda})|^2 \left| \frac{\partial}{\partial \nu_r} \frac{1}{h_\nu(\lambda)} \right| \text{sign}\left(\frac{\partial}{\partial \nu_r} \frac{1}{h_\nu(\lambda)}\right) \right) \Big|_{\nu=\nu_0} d\lambda \\ &\quad \times \int_{-\pi}^{\pi} \left(|g_\nu(e^{-i\lambda})|^2 \left| \frac{\partial}{\partial \nu_s} \frac{1}{h_\nu(\lambda)} \right| \text{sign}\left(\frac{\partial}{\partial \nu_s} \frac{1}{h_\nu(\lambda)}\right) \right) \Big|_{\nu=\nu_0} d\lambda \\ &= \text{Var}(\epsilon_0^2) \int_{-\pi}^{\pi} \left(|g_\nu(e^{-i\lambda})|^2 \frac{1}{h_\nu(\lambda)^2} \left| \frac{\partial}{\partial \nu_r} h_\nu(\lambda) \right| \text{sign}\left(\frac{\partial}{\partial \nu_r} h_\nu(\lambda)\right) \right) \Big|_{\nu=\nu_0} d\lambda \end{aligned}$$

$$\begin{aligned}
& \times \int_{-\pi}^{\pi} \left(|g_v(e^{-i\lambda})|^2 \frac{1}{h_v(\lambda)^2} \left| \frac{\partial}{\partial v_s} h_v(\lambda) \right| \operatorname{sign} \left(\frac{\partial}{\partial v_s} h_v(\lambda) \right) \right) \Big|_{v=v_0} d\lambda \\
& = \operatorname{Var}(\epsilon_0^2) \int_{-\pi}^{\pi} \frac{1}{h_v(\lambda)} \frac{\partial}{\partial v_r} h_v(\lambda) \Big|_{v=v_0} d\lambda \int_{-\pi}^{\pi} \frac{1}{h_v(\lambda)} \frac{\partial}{\partial v_s} h_v(\lambda) \Big|_{v=v_0} d\lambda \\
& = \operatorname{Var}(\epsilon_0^2) \frac{\partial}{\partial v_r} \int_{-\pi}^{\pi} \log h_v(\lambda) d\lambda \frac{\partial}{\partial v_s} \int_{-\pi}^{\pi} \log h_v(\lambda) d\lambda \Big|_{v=v_0} = 0.
\end{aligned}$$

This also means that cases (ii) and (iii) may include the terms with $m_1 = m_2 = n_1 = n_2$.

In case (ii) – that is, $m_1 = n_1 (= m) \neq m_2 = n_2 (= n)$ and including the case $m_1 = m_2 = n_1 = n_2$ as indicated above – we have

$$\begin{aligned}
W_{2,rs} &= (\mathbb{E}\epsilon_0^2)^2 \sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \ell_{1,v_0,-m}^{(r)} \ell_{2,v_0,-n}^{(r)} \ell_{1,v_0,k-m}^{(s)} \ell_{2,v_0,k-n}^{(s)} \\
&= (\mathbb{E}\epsilon_0^2)^2 \sum_{k=-\infty}^{\infty} \left(\sum_{m=-\infty}^{\infty} \ell_{1,v_0,-m}^{(r)} \ell_{1,v_0,k-m}^{(s)} \right) \left(\sum_{n=-\infty}^{\infty} \ell_{2,v_0,-n}^{(r)} \ell_{2,v_0,k-n}^{(s)} \right).
\end{aligned}$$

Note that the sum in the first parentheses is the convolution of $\{\ell_{1,v_0,-m}^{(r)}\}_{m \in \mathbb{Z}}$ and $\{\ell_{1,v_0,m}^{(s)}\}_{m \in \mathbb{Z}}$, and hence its Fourier transform is $\overline{\widehat{\ell}_{1,v_0}^{(r)}(\lambda)} \widehat{\ell}_{1,v_0}^{(s)}(\lambda)$. Similarly, the Fourier transform of the sequence in the second parentheses is $\overline{\widehat{\ell}_{2,v_0}^{(r)}(\lambda)} \widehat{\ell}_{2,v_0}^{(s)}(\lambda)$. By using Parseval's identity (A.1.3), we deduce that

$$\begin{aligned}
W_{2,rs} &= \frac{(\mathbb{E}\epsilon_0^2)^2}{2\pi} \int_{-\pi}^{\pi} \overline{\widehat{\ell}_{1,v_0}^{(r)}(\lambda)} \widehat{\ell}_{1,v_0}^{(s)}(\lambda) \overline{\widehat{\ell}_{2,v_0}^{(r)}(\lambda)} \widehat{\ell}_{2,v_0}^{(s)}(\lambda) d\lambda \\
&= 2\pi (\mathbb{E}\epsilon_0^2)^2 \int_{-\pi}^{\pi} \frac{1}{h_v(\lambda)} \frac{\partial}{\partial v_r} h_v(\lambda) \frac{1}{h_v(\lambda)} \frac{\partial}{\partial v_s} h_v(\lambda) \Big|_{v=v_0} d\lambda \\
&= 2\pi (\mathbb{E}\epsilon_0^2)^2 \int_{-\pi}^{\pi} \frac{\partial}{\partial v_r} \log h_v(\lambda) \frac{\partial}{\partial v_s} \log h_v(\lambda) \Big|_{v=v_0} d\lambda = 2\pi \sigma_0^4 c_{rs}(v_0).
\end{aligned}$$

Similarly, in case (iii) – that is, $m_1 = n_2 (= m) \neq m_2 = n_1 (= n)$ and including the case $m_1 = m_2 = n_1 = n_2$ as indicated above – we have

$$\begin{aligned}
W_{3,rs} &= (\mathbb{E}\epsilon_0^2)^2 \sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \ell_{1,v_0,-m}^{(r)} \ell_{2,v_0,-n}^{(r)} \ell_{1,v_0,k-n}^{(s)} \ell_{2,v_0,k-m}^{(s)} \\
&= (\mathbb{E}\epsilon_0^2)^2 \sum_{k=-\infty}^{\infty} \left(\sum_{m=-\infty}^{\infty} \ell_{1,v_0,-m}^{(r)} \ell_{2,v_0,k-m}^{(s)} \right) \left(\sum_{n=-\infty}^{\infty} \ell_{2,v_0,-n}^{(r)} \ell_{1,v_0,k-n}^{(s)} \right) \\
&= \frac{(\mathbb{E}\epsilon_0^2)^2}{2\pi} \int_{-\pi}^{\pi} \overline{\widehat{\ell}_{1,v_0}^{(r)}(\lambda)} \widehat{\ell}_{2,v_0}^{(s)}(\lambda) \overline{\widehat{\ell}_{2,v_0}^{(r)}(\lambda)} \widehat{\ell}_{1,v_0}^{(s)}(\lambda) d\lambda
\end{aligned}$$

and hence $W_{3,rs} = W_{2,rs} = 2\pi \sigma_0^4 c_{rs}(v_0)$. It follows that $W_{rs} = W_{1,rs} + W_{2,rs} + W_{3,rs} = 4\pi \sigma_0^4 c_{rs}(v_0)$.

This concludes the outline of the proof of the asymptotic normality (10.2.20). Proving the convergence (10.2.21) is not immediate – see the proof of Theorem 8.3.1, Giraitis et al. [406], pp. 216–218. The proof of the asymptotic normality (10.2.11), on the other hand, is

similar to that of (10.2.10) outlined above – see the proof of Theorem 8.3.1, Giraitis et al. [406], pp. 218–219. $\square$

Example 10.2.4 The FARIMA(0, d , 0) series have spectral density $f(\lambda) = \frac{\sigma^2}{2\pi} |1 - e^{-i\lambda}|^{-2d}$, corresponding to $v = d$, $g_v(e^{-i\lambda}) = (1 - e^{-i\lambda})^{-d}$ and $h_d(\lambda) = |1 - e^{-i\lambda}|^{-2d}$ in (10.2.6). Note that $\frac{\partial}{\partial d} \log h_d(\lambda) = -2 \log |1 - e^{-i\lambda}|$ and hence that

$$\begin{aligned} c(d) &= 4 \int_{-\pi}^{\pi} (\log |1 - e^{-i\lambda}|)^2 d\lambda = 8 \int_0^{\pi} \left(\log(2 \sin \frac{\lambda}{2}) \right)^2 d\lambda \\ &= 16 \int_0^{\pi} (\log(2 \sin u))^2 du = \frac{2\pi^3}{3}, \end{aligned}$$

where the last step follows by expanding the square $(\log(2 \sin u))^2 = (\log 2 + \log \sin u)^2$ and using the formulas 4.224.3 and 4.224.7 in Gradshteyn and Ryzhik [421], p. 531. It follows that the variance in the asymptotic result (10.2.10) for $\hat{d}_{whittle}$ is $4\pi(2\pi^3/3)^{-1} = 6/\pi^2$. See also Exercise 10.5.

Remark 10.2.5 The results presented in this section are asymptotic in nature, and apply to all considered estimation methods (exact MLE, Whittle approximation). How do the estimation methods compare for finite (small) samples? A nice and informative simulation study addressing this question can be found in Hauser [461]. See Section 10.9 for further discussion.

10.2.2 Autoregressive Approximation

Recall the expression (10.1.10) for the exact log-likelihood. The prediction $\hat{X}_j$ is based on the observations $X_{j-1}, X_{j-2}, \dots, X_1$. Consider first, however, the prediction $\hat{X}_j^\infty$ based on the infinite past $X_{j-1}, X_{j-2}, \dots$. The difference $X_j - \hat{X}_j^\infty$ between the true value X_j and its prediction $\hat{X}_j^\infty$ can be written as

$$X_j - \hat{X}_j^\infty = X_j - \sum_{k=1}^{\infty} \pi_{v,k} X_{j-k}, \quad (10.2.25)$$

where $\{\pi_{v,k}\}$ are defined formally through

$$1 - \sum_{k=1}^{\infty} \pi_{v,k} z^k = \frac{1}{\sum_{k=0}^{\infty} g_{v,k} z^k},$$

where $\{g_{v,k}\}$ is defined in (10.2.6). In other words, for the linear series $X_n = \sum_{j=0}^{\infty} g_{v,j} Z_{n-j}$, where $\{Z_n\}$ is a white noise series with variance $\sigma^2 = \sigma_Z^2$, the sequence $\{\pi_{v,k}\}$ appears in the inversion (*autoregressive representation*)

$$Z_n = X_n - \sum_{k=1}^{\infty} \pi_{v,k} X_{n-k}. \quad (10.2.26)$$

If the observations are extended to the infinite past, the prediction error would be given by

$$\mathbb{E}(X_j - \widehat{X}_j^\infty)^2 = \sigma^2. \quad (10.2.27)$$

The sequence $\{\pi_{v,k}\}$ could be computed efficiently and fast for many models of interest, for example, the FARIMA(p, d, q) family in Example 10.1.1.

Since the infinite past is not available in practice, we use instead $X_j - \widehat{X}_j$ and hence the approximate innovations $u_j(v) \approx X_j - \widehat{X}_j$, where

$$u_j(v) = X_j - \sum_{k=1}^{j-1} \pi_{v,k} X_{j-k}, \quad j = 1, \dots, N. \quad (10.2.28)$$

We also approximate the prediction errors v_{j-1} based on $\widehat{X}_j$ by σ^2 . Then, an approximation of the log-likelihood (10.1.10) based on autoregressive representation is

$$\ell_{AR}(\sigma^2, v) = \log \sigma^2 + \frac{1}{\sigma^2 N} \sum_{j=1}^N u_j(v)^2. \quad (10.2.29)$$

The corresponding ML estimators are defined as

$$(\widehat{\sigma}_{AR}^2, \widehat{v}_{AR}) = \underset{\sigma^2, v}{\operatorname{argmin}} \ell_{AR}(\sigma^2, v). \quad (10.2.30)$$

Note that $\widehat{v}_{AR}$ is obtained by minimizing the so-called conditional sum of squares $\sum_{j=1}^N u_j(v)^2$, and that

$$\widehat{\sigma}_{AR}^2 = \frac{1}{N} \sum_{j=1}^N u_j(\widehat{v}_{AR})^2. \quad (10.2.31)$$

One can show that, under suitable assumptions, the estimators $\widehat{\sigma}_{AR}^2, \widehat{v}_{AR}$ satisfy the same asymptotic normality results as in Theorem 10.2.1 (Beran [122] and Section 5.5.5 in Beran et al. [127]). It should also be noted that these estimators can be applied to nonstationary series (Hualde and Robinson [498]).

10.3 Model Selection and Diagnostics

The estimation methods considered in Sections 10.1 and 10.2 assume that the model (order) is known (e.g., p and q in the FARIMA(p, d, q) class are known). In order to avoid overfitting, it is common to choose a model (order) using an information criterion wherein one minimizes

$$2\widehat{\ell} + \alpha s, \quad (10.3.1)$$

where $\widehat{\ell}$ denotes the (non-scaled¹) negative log-likelihood evaluated at the estimated parameter values, and s is the number of parameters used (e.g., $p + q + 2$ for the FARIMA(p, d, q))

¹ By “non-scaled” we mean not divided by $N/2$ as in (10.1.3).

class, where 2 counts the parameters σ^2 and d). The value α in (10.3.1) depends on the criterion used:

$$\begin{aligned} \text{AIC: } & \alpha = 2 \\ \text{BIC: } & \alpha = 2 \log N \\ \text{HIC: } & \alpha = c \log \log N \end{aligned} \quad (10.3.2)$$

with $c > 1$ (AIC for Akaike Information Criterion, Akaike [17]; BIC for Bayesian Information Criterion, Schwarz [894]; HIC for Hannan Information Criterion, Hannan and Quinn [452]). The negative log-likelihood can be an approximate one as in (10.2.29), except that the latter should not be scaled by $N/2$ since (10.3.1) involves a non-scaled log-likelihood. Indeed, note that (10.2.29) is an approximation to (10.1.10) or, equivalently, (10.1.3). Thus, the approximate negative log-likelihood (10.2.29) needs to be multiplied by $N/2$; that is, the negative log-likelihood in (10.3.1) can be approximated by

$$\widehat{\ell} = \frac{N}{2} \ell_{AR}(\widehat{\sigma}_{AR}^2, \widehat{v}_{AR}) = \frac{N}{2} \log \widehat{\sigma}_{AR}^2 + \frac{N}{2}, \quad (10.3.3)$$

where we used (10.2.31). (Note that the last term $\frac{N}{2}$ can be ignored in minimizing (10.3.1).) The basic idea behind (10.3.1) is that the first term $2\widehat{\ell}$ decreases with the increasing number of parameters, whereas the second term αs acts as a penalty, which increases with the increasing number of parameters.

Some of the criteria above have been investigated for certain classes of LRD models. An empirical study of AIC in FARIMA(p, d, q) class was carried out in Crato and Ray [269]. Theoretical properties of AIC, BIC and HIC in FARIMA($p, d, 0$) class and the choice (10.3.3) were studied in Beran, Bhansali, and Ocker [124]. In particular, the order $\widehat{p}$ chosen by BIC and HIC was shown to be consistent, whereas the order chosen by AIC overestimates the true p with positive probability. A more recent work on the model selection can be found in Baillie, Kapetanios, and Papailias [90].

Once a time series model is chosen, its goodness of fit should be assessed through diagnostics checks. As with other statistical models, it is common to examine the normalized residuals. In the time series context, these are defined as

$$\widehat{\epsilon}_t = \frac{X_t - \widehat{X}_t(\widehat{\eta})}{v_{t-1}^{1/2}(\widehat{\eta})}, \quad (10.3.4)$$

where $\widehat{X}_t$ and v_{t-1} are defined as in (10.1.10) and evaluated at the estimated parameter η value. As in (10.2.28)–(10.2.31), the residuals could also be approximated through

$$\widehat{\epsilon}_t \approx \frac{u_t(\widehat{v}_{AR})}{\widehat{\sigma}_{AR}}. \quad (10.3.5)$$

If the model fits the data well, the residuals should be consistent with the white noise assumption. In the case of short-range dependent series, this is often assessed by examining the residual plot, and the portmanteau tests for residuals (Box–Pierce, Ljung–Box and others) based on the sample autocorrelation function coefficients of the residuals (see, for example, Brockwell and Davis [186]).

In the case of LRD series, these common diagnostics checks are problematic. The residuals (10.3.4) may be tedious to compute, especially for longer time series. The classical

portmanteau tests involve only the sample autocorrelation function coefficients at a number of small lags, whereas LRD is intrinsically related to the autocorrelation function at large lags. A number of goodness-of-fit tests for LRD series were constructed to deal with these issues.

Chen and Deo [220] considered the test statistic

$$T_N(\hat{\eta}) = \frac{\frac{2\pi}{N} \sum_{j=0}^{N-1} \tilde{f}_e(\hat{\eta}, \lambda_j)^2}{\left(\frac{2\pi}{N} \sum_{j=0}^{N-1} \tilde{f}_e(\hat{\eta}, \lambda_j) \right)^2}, \quad (10.3.6)$$

where

$$\tilde{f}_e(\hat{\eta}, \lambda) = \frac{1}{N} \sum_{k=1}^{N-1} \frac{W(\lambda - \lambda_k) I_X(\lambda_k)}{f(\hat{\eta}, \lambda_k)}, \quad (10.3.7)$$

$I_X(\lambda)$ is the periodogram,² $\lambda_j = \frac{2\pi k}{N}$ are the Fourier frequencies, $\hat{\eta}$ is an estimator of η , $\{f(\eta, \lambda)\}$ is a considered class of spectral densities (e.g., FARIMA(p, d, q) class) and W is the spectral window defined by

$$W(\lambda) = \frac{1}{2\pi} \sum_{|h| < N} k\left(\frac{h}{p_N}\right) e^{-ih\lambda},$$

where k is a kernel function (e.g., Bartlett $k(z) = (1 - |z|)1_{[-1,1]}(z)$) and p_N is a bandwidth. Under suitable assumptions for both short-range and long-range dependent series, Chen and Deo [220] showed that, as $N \rightarrow \infty$,

$$\frac{N(T_N(\hat{\eta}) - C_N(k))}{D_N(k)^{1/2}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (10.3.8)$$

where

$$C_N(k) = \frac{1}{N\pi} \sum_{j=1}^{N-1} \left(1 - \frac{j}{N}\right) k^2\left(\frac{j}{p_N}\right) + \frac{1}{2\pi},$$

$$D_N(k) = \frac{2}{\pi^2} \sum_{j=1}^{N-2} \left(1 - \frac{j}{N}\right) \left(1 - \frac{j+1}{N}\right) k^4\left(\frac{j}{p_N}\right).$$

The asymptotic result (10.3.8) is used to devise a formal test where the null hypothesis is that the time series has the spectral density in the class of interest. One rejects the null hypothesis at a significance level α when the test statistic (the expression on the left-hand side of (10.3.8)) exceeds the critical value z_α , where z_α satisfies $\mathbb{P}(\mathcal{N}(0, 1) > z_\alpha) = \alpha$. Note that the test statistic (10.3.6) is not based on the model residuals and is easy to compute. It can nevertheless be viewed as a generalized portmanteau statistic involving *all* sample autocorrelation function coefficients of the residuals (see the discussion on pp. 384–385 in Chen and Deo [220]).

² Our definition of the periodogram differs by the factor 2π from that in Chen and Deo [220] – see also the warning in Section 1.3.3.

A statistic similar to (10.3.6) was also considered in Beran [120], with the correction in Deo and Chen [303]. Other testing procedures, based on the periodogram as well, can be found in Fay and Philippe [355], Delgado, Hidalgo, and Velasco [300].

10.4 Forecasting

A common objective of a time series analysis is (linear) forecasting (or prediction).³ What about forecasting long-range dependent time series? These series are second-order stationary, and can be forecast in the general framework of second-order stationary series (e.g., Brockwell and Davis [186]). For example, a one-step forecast is defined as

$$\hat{X}_{N+1} = \varphi_{N1}X_N + \varphi_{N2}X_{N-1} + \cdots + \varphi_{NN}X_1, \quad (10.4.1)$$

where the coefficients φ_{Ni} can be obtained from the Durbin–Levinson algorithm (cf. (10.1.4)) and the autocovariance function obtained from the fitted model. Similarly, a two-step forecast is defined by⁴

$$\hat{X}_{N+2} = \varphi_{N+1,1}\hat{X}_{N+1} + \varphi_{N+1,2}X_N + \cdots + \varphi_{N+1,N+1}X_1, \quad (10.4.2)$$

that is, as (10.4.1) but replacing the unobserved X_{N+1} by its forecast $\hat{X}_{N+1}$, and so on. The mean-squared error of the one-step forecast is $v_N = \mathbb{E}(X_{N+1} - \hat{X}_{N+1})^2$ (cf. (10.1.5)). It can be used to construct an approximate Gaussian confidence interval associated with the forecast $\hat{X}_{N+1}$.

Though forecasting long-range dependent series fits in the general framework of second-order stationary series, there is also one natural feature that sets it apart. To explain this point, suppose N is large and consider a one-step forecast $\hat{X}_{N+1}$ in (10.4.1). As in (10.2.26), we could write

$$X_{N+1} = \sum_{k=1}^{\infty} \pi_k X_{N+1-k} + Z_{N+1}. \quad (10.4.3)$$

If the forecast $\hat{X}_{N+1}^{\infty}$ were based on the infinite past $X_N, X_{N-1}, \dots$, then

$$\hat{X}_{N+1}^{\infty} = \sum_{k=1}^{\infty} \pi_k X_{N+1-k}. \quad (10.4.4)$$

If N is assumed large, it is then expected that

$$\varphi_{Nk} \approx \pi_k. \quad (10.4.5)$$

On the other hand, for many LRD models with parameter $d \in (0, 1/2)$ (e.g., FARIMA(0, d , 0) model for which $(I - B)^{-d}X_n = Z_n$), we have for large k ,

$$\pi_k \approx Ck^{-d-1}.$$

Hence, for large N ,

$$\varphi_{Nk} \approx Ck^{-d-1}. \quad (10.4.6)$$

³ We do not distinguish between the terms “forecast” and “prediction.” See also Section 6.4.2.

⁴ We write $\varphi_{N+1,1}$ instead of $\varphi_{N+1,1}$ and so on to avoid possible confusion.

This shows that, for LRD series, the influence of the past observations decays as a power function if measured through the coefficients φ_{Nk} .

In fact, the view presented above draws a somewhat simplified picture. For example, as stated in Exercise 2.8, for FARIMA(0, d , 0) series,

$$\varphi_{NN} = \frac{d}{N-d} \sim \frac{d}{N}, \quad (10.4.7)$$

as $N \rightarrow \infty$. The same asymptotic behavior holds also for the entire class of long-range dependent FARIMA(p, d, q) models (Inoue [519]). Note that the decay in (10.4.7) is even slower than that in (10.4.6), with k replaced by N .

10.5 R Packages and Case Studies

We illustrate some of the methods presented above on real time series using R (R Core Team [838]). Several R packages are available to work with long-range dependent time series. We consider two of them below: `arfima` [976] and `fracdiff` [782], along with the use of the package `forecast` [1008] for the latter.

10.5.1 The ARFIMA Package

We consider first the `arfima` package. Figure 10.1 presents the time plot of the series `tmpyr` available in the package, and its sample autocorrelation function (ACF). The `tmpyr` series represents the central England mean yearly temperatures from 1659 to 1976. The number of observations is $N = 318$. The plots are obtained through the functions:

```
data(tmpyr)
plot(tmpyr)
acf(tmpyr, max.lag=100)
```

(The function `acf` is part of the general R stats package.)

Table 10.1 lists the AIC values for a number of fitted FARIMA(p, d, q) models. We focus on AIC (as opposed to other information criteria) since for the temperature time series, it

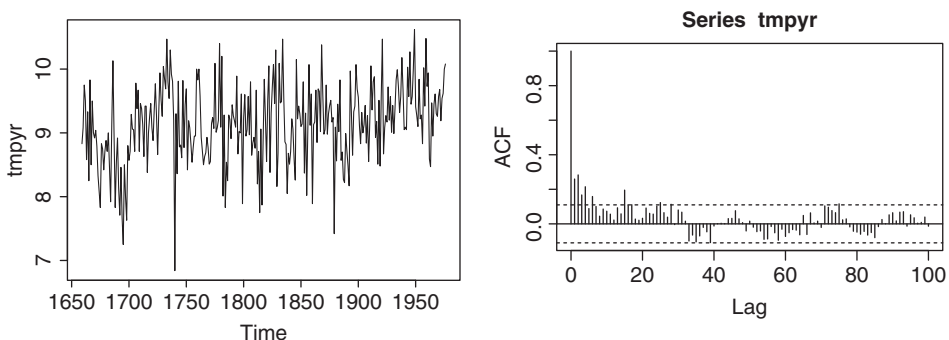


Figure 10.1 The time plot and the sample ACF of the central England mean yearly temperatures from 1659 to 1976.

Table 10.1 The AIC values for the listed models.

ARMA(<i>p</i> , <i>q</i>)			FARIMA(<i>p</i> , <i>d</i> , <i>q</i>)		
<i>p</i>	<i>q</i>	AIC	<i>p</i>	<i>q</i>	AIC
			0	0	−337.83
1	0	−319.56	1	0	−338.98
0	1	−312.18	0	1	−323.21
1	1	−337.99	1	1	−339.75
2	0	−335.37	2	0	−338.47
0	2	−326.26	0	2	−337.76

was considered in some early works on long-range dependence, e.g., Hosking [485]. For example, the AIC value for the FARIMA(1, *d*, 1) model is obtained through:

```
fit_1d1 <- arfima(tmpyr,order = c(1,0,1), numeach = c(2,1), dmean = FALSE)
fit_1d1
AIC(fit_1d1)
```

(the input `order = c(1, 0, 1)` sets the FARIMA(1, *d*, 1) model, the option `numeach = c(2, 1)` sets the number of starting values for each parameter in the numerical optimization where the 2 refers to the ARMA parameters, the 1 to the long-range dependence parameter, and the option `dmean = FALSE` corresponds to the centered series where the maximization does not involve the mean). It is seen from the table (the entry in bold) that FARIMA(1, *d*, 1) is the model with the lowest AIC value. The model FARIMA(1, *d*, 0) has the second lowest AIC. The estimated parameter values (and the respective standard errors in parentheses) for the FARIMA(1, *d*, 1) fit are: $\hat{d} = 0.278$ (0.05), $\hat{\phi}_1 = -0.748$ (0.147), $\hat{\theta}_1 = 0.647$ (0.173),⁵ $\hat{\sigma}^2 = 0.335$ and the sample mean $\hat{\mu} = 9.143$.

Figure 10.2 presents the prediction values and confidence intervals of the `tmpyr` series for five steps ahead based on the FARIMA(1, *d*, 1) model. The prediction and the plot is obtained through:

```
pred <- predict(fit_1d1, n.ahead = 5, bootpred = FALSE)
pred
plot(pred, numback = 25)
```

(the option `bootpred = FALSE` is to not produce bootstrap prediction and intervals, and the option `numback = 25` sets the number of the past observations of the original series to include in the plot). Two starting values were used for the likelihood when fitting the FARIMA(1, *d*, 1) model (see above). The top plot in Figure 10.2 corresponds to the starting value with the lowest AIC value and the bottom plot to the next lowest AIC value.

10.5.2 The *FRACDIFF* Package

We now turn to the `fracdiff` package. The function `fracdiff` in the package is one of the commonly used R functions to fit a FARIMA(*p*, *d*, *q*) model. Figure 10.3 presents the

⁵ The convention used in the package has a different sign for the MA parameters θ . The numerical value given here follows our convention (see Example 2.3.3).

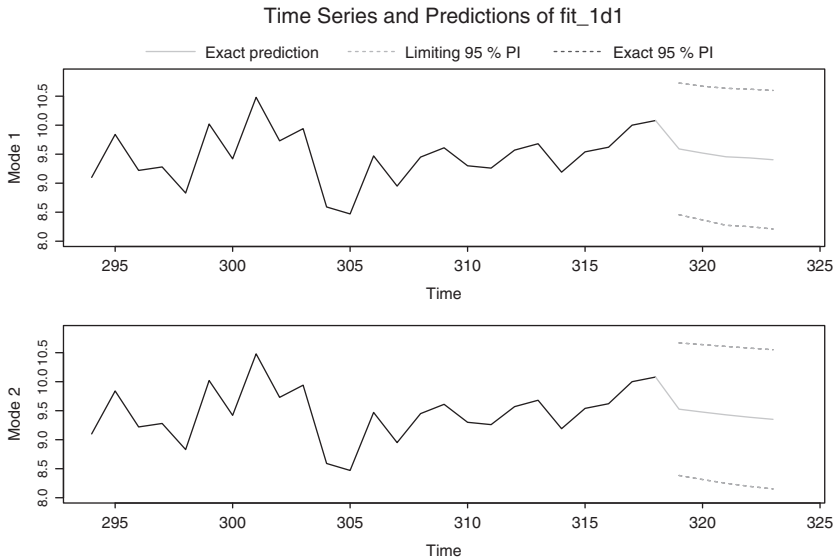


Figure 10.2 The five-step-ahead prediction of the central England mean yearly temperatures.

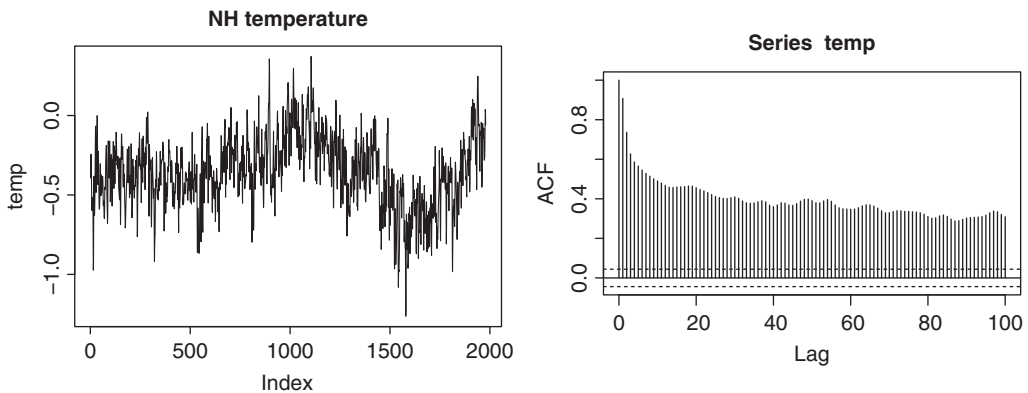


Figure 10.3 The time plot and the sample ACF of the Northern Hemisphere temperatures from 1 to 1979AD.

reconstructed annual NH (Northern Hemisphere) temperatures from 1 to 1979AD, analyzed by Moberg, Sonechkin, Holmgren, Datsenko, and Karlen [724], Mills [721].

Table 10.2 lists the AIC values for several FARIMA(p, d, q) models fitted to the Northern Hemisphere temperature data. For example, the entry for the FARIMA(2, d , 1) model is obtained through:

```
fit_2d1 <- fracdiff(temp - mean(temp), nar = 2, nma = 1)
summary(fit_2d1)
```

The lowest AIC value is obtained for the FARIMA(2, d , 2) model. The corresponding parameter estimates (and their standard errors in parentheses) are: $\hat{d} = 0.44$ (0.02), $\hat{\varphi}_1 = 0.32$ (0.03), $\hat{\varphi}_2 = -0.25$ (0.025), $\hat{\theta}_1 = 1.51$ (0.03), $\hat{\theta}_2 = 0.603$ (0.025), together with the sample mean $\hat{\mu} = -0.3537$ and the estimated standard deviation of the innovations $\hat{\sigma} = 0.0484$.

Table 10.2 The AIC values for the listed models.

FARIMA(p, d, q)			FARIMA(p, d, q)		
p	q	AIC	p	q	AIC
0	0	−3196.20	2	2	−6353.06
1	0	−4156.66	3	1	−6265.40
0	1	−5215.21	1	3	−6293.05
1	1	−5696.02	4	0	−5863.42
2	0	−5044.89	0	4	−6347.40
0	2	−6238.31	3	2	−6345.39
2	1	−6153.44	2	3	−6307.86
1	2	−5696.02	4	1	−6329.35
3	0	−6292.18	1	4	−6342.41
0	3	−5489.88			

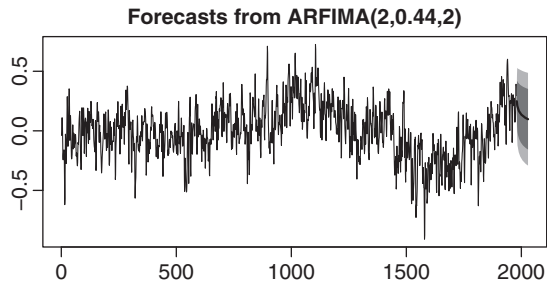


Figure 10.4 The five-step-ahead prediction of the Northern Hemisphere temperatures is depicted in the shaded region at the end of the graph.

Finally, Figure 10.4 depicts the forecasts of the demeaned time series for five steps ahead based on the fitted FARIMA(2, d , 2) model. The plot is obtained through `plot(forecast(fit_2d2,h=50))`. The darker shaded band in the forecast region at the end of the graph corresponds to 80% confidence band. The wider band in lighter shade corresponds to 90% confidence band.

10.6 Local Whittle Estimation

Exact or approximate MLE is based on the possibly optimistic assumption that a specific model is correct. Moreover, for a parametric family of models with increasing number of parameters such as FARIMA(p, d, q) series in Example 10.1.1, MLE together with a model selection criterion will tend to favor models in the parametric family with fewer parameters. If one is specifically interested in the LRD parameter, one could use semi-parametric estimation, where the spectral density is specified around the frequency zero only. For example, the GPH estimator discussed in Section 2.10.3 is an example of a semi-parametric estimator. We discuss here another popular and more efficient semi-parametric estimator, called the *local Whittle estimator*.

10.6.1 Local Whittle Estimator

Recall the discrete Whittle approximation of the log-likelihood in (10.2.5). The idea of *local Whittle estimation* is to use the lowest m Fourier frequencies only, and consider

$$\ell_{lw}(\eta) = \frac{1}{m} \sum_{j=1}^m \left(\log f_{\eta} \left(\frac{2\pi j}{N} \right) + \frac{\tilde{I}_X \left(\frac{2\pi j}{N} \right)}{f_{\eta} \left(\frac{2\pi j}{N} \right)} \right), \quad (10.6.1)$$

where we excluded the term with $\log(2\pi)$ resulting from the first sum of (10.2.5) and denote the periodogram normalized by 2π as

$$\tilde{I}_X(\lambda) = \frac{I_X(\lambda)}{2\pi} \quad (10.6.2)$$

for notational convenience throughout this section (see also the warning in Section 1.3.3). Assuming $f_{\eta}(\lambda) \sim c\lambda^{-2d}$, $0 < d < 1/2$, and $m/N \rightarrow 0$, the minimization of $\ell_{lw}(\eta)$ can be replaced by the minimization of

$$Q(c, d) = \frac{1}{m} \sum_{j=1}^m \left(\log(c\lambda_j^{-2d}) + \frac{\tilde{I}_X(\lambda_j)}{c\lambda_j^{-2d}} \right), \quad (10.6.3)$$

where $\lambda_j = \frac{2\pi j}{N}$. The parameter vector η of the spectral density $f_{\eta}(\lambda)$ now consists of the parameters c, d .

Differentiating with respect to c leads to

$$\frac{\partial Q}{\partial c}(c, d) = \frac{1}{m} \sum_{j=1}^m \left(c^{-1} - \frac{\tilde{I}_X(\lambda_j)}{c^2 \lambda_j^{-2d}} \right).$$

Setting this to zero gives

$$\hat{c} = \frac{1}{m} \sum_{j=1}^m \frac{\tilde{I}_X(\lambda_j)}{\lambda_j^{-2d}} =: G(d).$$

Then, the *local Whittle estimator* of d is defined as

$$\hat{d}_{lw} = \operatorname{argmin}_{d \in \Theta} Q(d), \quad (10.6.4)$$

where $\Theta \subset (-1/2, 1/2)$ and

$$Q(d) = \log G(d) - d \frac{2}{m} \sum_{j=1}^m \log \lambda_j = \log \left(\frac{1}{m} \sum_{j=1}^m \frac{\tilde{I}_X(\lambda_j)}{\lambda_j^{-2d}} \right) - d \frac{2}{m} \sum_{j=1}^m \log \lambda_j, \quad (10.6.5)$$

that is, by focusing on the first term in (10.6.3) and replacing c by $G(d)$.

It is useful to indicate first why the local Whittle estimator is expected to estimate the true parameter d_0 . Assuming $\tilde{I}_X(\lambda_j) \approx f_{\eta_0}(\lambda_j) \sim c_0 \lambda_j^{-2d_0}$, the objective function in (10.6.5) is approximately

$$\begin{aligned} \log \left(\frac{1}{m} \sum_{j=1}^m c_0 \lambda_j^{-2(d_0-d)} \right) - d \frac{2}{m} \sum_{j=1}^m \log \lambda_j &= R \left(\frac{m}{N}, c_0, d_0 \right) \\ &+ \log \left(\frac{1}{m} \sum_{j=1}^m \left(\frac{j}{m} \right)^{-2(d_0-d)} \right) - d \frac{2}{m} \sum_{j=1}^m \log \frac{j}{m} \end{aligned}$$

$$\begin{aligned} &\approx R\left(\frac{m}{N}, c_0, d_0\right) + \log \int_0^1 x^{-2(d_0-d)} dx - 2d \int_0^1 \log x dx \\ &= R\left(\frac{m}{N}, c_0, d_0\right) - \log(1 - 2(d_0 - d)) + 2d =: \overline{Q}(d), \end{aligned}$$

where R is some function that does not depend on d . Note that the function $\overline{Q}(d)$ is minimized at $d = d_0$.

In the following result, we provide the asymptotic normality result for the local Whittle estimator $\widehat{d}_{lw}$, due to Robinson [853]. The following assumptions will be made:

(B1) For some $\rho \in (0, 2]$, the spectral density satisfies

$$f_{\eta_0}(\lambda) = c_0 \lambda^{-2d_0} (1 + O(\lambda^\rho)), \quad (10.6.6)$$

as $\lambda \rightarrow 0$, where $c > 0$ and $d_0 \in [\Delta_1, \Delta_2]$ ($-1/2 < \Delta_1 < \Delta_2 < 1/2$).

(B2) The series satisfies

$$X_n - \mathbb{E}X_0 = \sum_{j=0}^{\infty} \alpha_j \epsilon_{n-j}, \quad \sum_{j=0}^{\infty} \alpha_j^2 < \infty, \quad (10.6.7)$$

where

$$\mathbb{E}(\epsilon_n | \mathcal{F}_{n-1}) = 0, \quad \mathbb{E}(\epsilon_n^2 | \mathcal{F}_{n-1}) = 1, \quad \mathbb{E}(\epsilon_n^3 | \mathcal{F}_{n-1}) = \mu_3 \text{ a.s.}, \quad \mathbb{E}\epsilon_n^4 = \mu_4,$$

for finite constants μ_3, μ_4 and $\mathcal{F}_n = \sigma\{\epsilon_k, k \leq n\}$. Moreover, there exists a random variable ϵ such that $\mathbb{E}\epsilon^2 < \infty$ and $\mathbb{P}(|\epsilon_n| > x) \leq K\mathbb{P}(|\epsilon| > x)$ for some $K > 0$ and all $x \in \mathbb{R}$.

(B3) In a neighborhood of zero, $\alpha(\lambda) = \sum_{j=0}^{\infty} \alpha_j e^{-ij\lambda}$ is differentiable, and

$$\frac{d}{d\lambda} \alpha(\lambda) = O\left(\frac{|\alpha(\lambda)|}{\lambda}\right), \quad \text{as } \lambda \rightarrow 0.$$

(B4) As $N \rightarrow \infty$,⁶

$$\frac{1}{m} + \frac{m^{1+2\rho}}{N^{2\rho}} \rightarrow 0. \quad (10.6.8)$$

Theorem 10.6.1 *Let $\widehat{d}_{lw}$ be the local Whittle estimator of d defined in (10.6.4). Under Assumptions (B1)–(B4) above,*

$$\sqrt{m}(\widehat{d}_{lw} - d_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{4}\right). \quad (10.6.9)$$

Remark 10.6.2 Note the slower rate of convergence $\sqrt{m}$ when compared to $\sqrt{N}$ in (10.2.10). Note also the gain in efficiency when compared to the GPH estimator in (2.10.18). Here the asymptotic variance is $1/4$ whereas for the GPH it is $\pi^2/24 = 0.41$, which is greater. Thus, the gain in efficiency is $(1/4)/(\pi^2/24) = 1.64$.

⁶ The corresponding Assumption (A4') in Robinson [853], p. 1641, carries a factor $(\log m)^2$ next to $m^{1+2\rho}$ in (10.6.8). This factor is not needed as shown by Andrews and Sun [34], p. 577.

Proof outline of Theorem 10.6.1 We provide here an outline of the proof of the theorem. A complete proof can be found in Robinson [853], Theorem 2. See also Theorem 2 and Corollary 1 in Andrews and Sun [34] for a slightly different approach in a more general context.

The first step in the proof is to show that $\widehat{d}_{lw}$ is consistent. We will not discuss the proof of consistency here (see Theorem 1 in Robinson [853], or Andrews and Sun [34] for a different approach). In view of (10.6.4)–(10.6.5), the first order condition and the Taylor's expansion yield

$$0 = \frac{\partial Q}{\partial d}(\widehat{d}_{lw}) = \frac{\partial Q}{\partial d}(d_0) + \frac{\partial^2 Q}{\partial^2 d}(d^*)(\widehat{d}_{lw} - d_0),$$

where $|d^* - d_0| \leq |\widehat{d}_{lw} - d_0|$. Hence,

$$\sqrt{m}(\widehat{d}_{lw} - d_0) = -\frac{1}{\frac{\partial^2 Q}{\partial^2 d}(d^*)} \sqrt{m} \frac{\partial Q}{\partial d}(d_0). \quad (10.6.10)$$

It is then enough to show that $\sqrt{m} \frac{\partial Q}{\partial d}(d_0) \xrightarrow{d} \mathcal{N}(0, 4)$ and $\frac{\partial^2 Q}{\partial^2 d}(d^*) \xrightarrow{p} 4$.

Note that (10.6.5) implies

$$\frac{\partial Q}{\partial d}(d_0) = \left(\frac{2}{m} \sum_{j=1}^m \log \lambda_j \frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} \right) \left(\frac{1}{m} \sum_{j=1}^m \frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} \right)^{-1} - \frac{2}{m} \sum_{j=1}^m \log \lambda_j \quad (10.6.11)$$

$$= - \left[\frac{1}{m} \sum_{j=1}^m (z_j - \bar{z}) \left(\frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} - 1 \right) \right] \left(\frac{1}{m} \sum_{j=1}^m \frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} \right)^{-1}, \quad (10.6.12)$$

where $z_j = -2 \log \lambda_j$ and $\bar{z} = \frac{1}{m} \sum_{j=1}^m z_j$, and we could add c_0 in (10.6.11) since it cancels out after multiplication with its reciprocal. In view of (2.10.15), one expects that

$$\frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} \approx \xi_j, \quad (10.6.13)$$

where ξ_j are independent $\chi^2_2/2$ (or standard exponential) random variables. Since

$$\frac{1}{m} \sum_{j=1}^m \frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} \approx \frac{1}{m} \sum_{j=1}^m \xi_j \approx \mathbb{E} \xi_j = 1,$$

we have

$$\sqrt{m} \frac{\partial Q}{\partial d}(d_0) \approx -\frac{1}{\sqrt{m}} \sum_{j=1}^m (z_j - \bar{z})(\xi_j - 1). \quad (10.6.14)$$

By the Lindeberg–Feller central limit theorem for independent, non-identically distributed random variables (Feller [360], Chapter VIII, Section 4), the right-hand side converges in distribution to $\mathcal{N}(0, \sigma^2)$, where

$$\sigma^2 = \lim_{m \rightarrow \infty} \frac{1}{m} \sum_{j=1}^m (z_j - \bar{z})^2 = 4,$$

by using Proposition 2.10.2.

Similarly, from (10.6.11),

$$\begin{aligned} \frac{\partial^2 Q}{\partial d^2}(d) &= 4 \frac{\left(\frac{1}{m} \sum_{j=1}^m (\log \lambda_j)^2 \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right) \left(\frac{1}{m} \sum_{j=1}^m \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right) - \left(\frac{1}{m} \sum_{j=1}^m \log \lambda_j \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right)^2}{\left(\frac{1}{m} \sum_{j=1}^m \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right)^2} \\ &= 4 \frac{\left(\frac{1}{m} \sum_{j=1}^m (\log j)^2 \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right) \left(\frac{1}{m} \sum_{j=1}^m \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right) - \left(\frac{1}{m} \sum_{j=1}^m \log j \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right)^2}{\left(\frac{1}{m} \sum_{j=1}^m \frac{\tilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d}}\right)^2}, \end{aligned}$$

where the last equality follows by simple calculation using $\lambda_j = 2\pi j/N$. Since d^* approaches d_0 in probability and using the approximation (10.6.13), it is expected that

$$\frac{\partial^2 Q}{\partial d^2}(d^*) \approx 4 \frac{\left(\frac{1}{m} \sum_{j=1}^m (\log j)^2 \xi_j\right) \left(\frac{1}{m} \sum_{j=1}^m \xi_j\right) - \left(\frac{1}{m} \sum_{j=1}^m (\log j) \xi_j\right)^2}{\left(\frac{1}{m} \sum_{j=1}^m \xi_j\right)^2}. \quad (10.6.15)$$

Since $\text{Var}(m^{-1} \sum_{j=1}^m (\log j)^k \xi_j) = Cm^{-2} \sum_{j=1}^m (\log j)^{2k} \sim Cm^{-2} \int_1^m (\log x)^{2k} dx \sim Cm^{-1} (\log m)^{2k}$, for $k = 0, 1, 2$, we have

$$\frac{1}{m} \sum_{j=1}^m (\log j)^k \xi_j - \frac{1}{m} \sum_{j=1}^m (\log j)^k = O_p\left(\frac{(\log m)^k}{m^{1/2}}\right) \rightarrow 0,$$

as $m \rightarrow \infty$, so we can approximate the first term on the left-hand side by the second term. We also have $m^{-1} \sum_{j=1}^m \xi_j \rightarrow \mathbb{E} \xi_j = 1$ as $m \rightarrow \infty$. Hence, the right-hand side of (10.6.15) behaves in probability as

$$4 \left(\frac{1}{m} \sum_{j=1}^m (\log j)^2 - \left(\frac{1}{m} \sum_{j=1}^m \log j \right)^2 \right) = 4 \frac{1}{m} \sum_{j=1}^m \left(\log j - \frac{1}{m} \sum_{j=1}^m \log j \right)^2 = \frac{1}{m} \sum_{j=1}^m (z_j - \bar{z})^2 \rightarrow 4,$$

where $z_j = -2 \log \frac{2\pi j}{N}$ and using Proposition 2.10.2 for the last convergence. $\square$

In practice, the local Whittle estimation is accompanied by a plot of the estimated values $\hat{d}_{lw}$ against the number of lower frequencies m used. Such plot is referred to as the (local) *Whittle plot*. Since the numerical optimization procedure can find minimizers d outside the primary range $(-1/2, 1/2)$, especially when m is small, the vertical axis in the Whittle plot is often larger than this range to indicate such values of d . For example, the left plot of Figure 10.5 presents the Whittle plot for a Gaussian FARIMA(0, 0.4, 0) series of length $N = 1024$. The 95% asymptotic confidence bands, based on Theorem 10.6.1, are also plotted in dash-dot lines. The right plot of the figure presents the Whittle plots for 20 realizations of the series.

10.6.2 Bandwidth Selection

The estimated values $\hat{d}_{lw}$ in the Whittle plots of Figure 10.5 are close to the true value $d = 0.4$ for all numbers of lower frequencies m used. This is not necessarily a typical behavior.

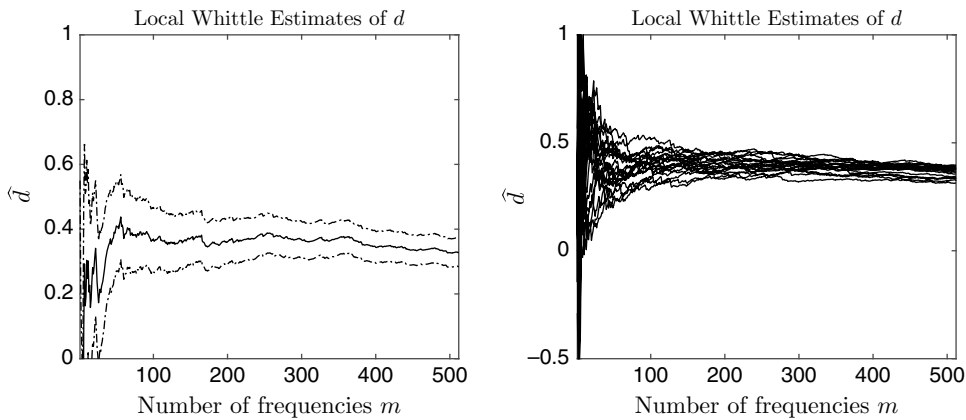


Figure 10.5 Local Whittle plots for Gaussian FARIMA(0, 0.4, 0) series of length $N = 1024$. Left: one realization with confidence bands. Right: 20 realizations.

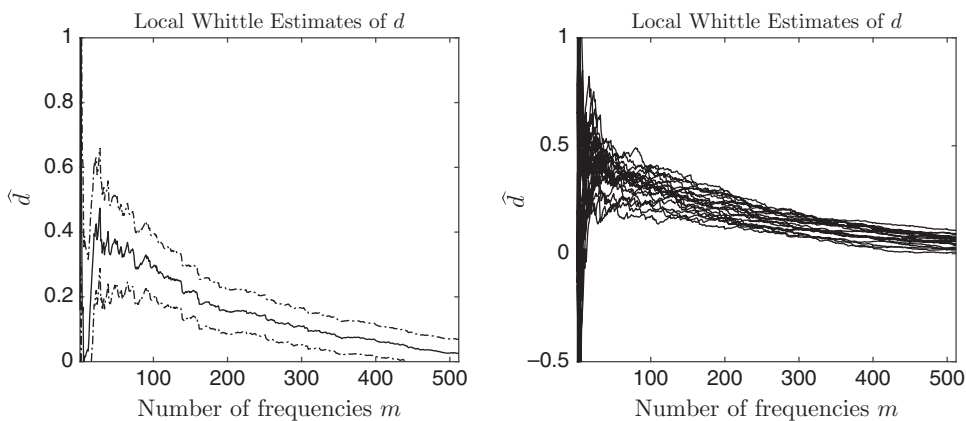


Figure 10.6 Local Whittle plots for Gaussian FARIMA(0, d , 1) series of length $N = 1024$ with $d = 0.4$ and $\theta_1 = -2$. Left: one realization with confidence bands. Right: 20 realizations.

For example, Figure 10.6 presents similar Whittle plots for a Gaussian FARIMA(0, d , 1) series with $d = 0.4$ and the MA parameter $\theta_1 = -2$. The estimated value $\hat{d}_{lw}$ is now much more sensitive to the choice of the number of frequencies m . The value m is referred to as the *bandwidth*. How does one choose the bandwidth in such situations?

The reason that the choice of the bandwidth is not immediate, for example, in Figure 10.6, is two-fold. On one hand, for larger bandwidth m , the spectral density may deviate from the power law behavior stipulated by the model. This translates into a bias of the local Whittle estimator, as in Figure 10.6. On the other hand, for smaller m , no bias is expected but the variance of the local Whittle estimator is now larger. This phenomenon is quite typical in many estimation problems and is known as the *bias-variance tradeoff*.

A bandwidth m should be chosen to balance the bias-variance tradeoff. If the decision on m needs to be made from a single time series realization and hence from one single Whittle plot, a basic idea is to choose a region of m where the plot “flattens out” for smaller values

of m and the variance is not “too large” yet. In the left plot of Figure 10.6, this would be a rough range of m from 25 to 75.

The bias-variance tradeoff could also be examined in more mathematical terms, leading to plug-in (automatic) choices of the bandwidth if such are desired (e.g., when dealing with many realizations of time series). More specifically, suppose that the spectral density of the time series satisfies

$$f_{\eta_0}(\lambda) = c_0 \lambda^{-2d_0} (1 + c_\rho \lambda^\rho + o(\lambda^\rho)), \quad \rho > 0, \quad (10.6.16)$$

as $\lambda \rightarrow 0$. The value $\rho = 2$ refers to the so-called *smooth* models such as FARIMA series. *Non-smooth* models are also of interest. For example, consider the time series model $X_n + Y_n$, where X_n has the spectral density $c_0 \lambda^{-2d_0}$ and Y_n is a white noise. (Similar models are considered in the context of stochastic volatility.) Then, the spectral density of $X_n + Y_n$ is $c_0 \lambda^{-2d_0} + \sigma_Y^2/(2\pi)$ and hence satisfies (10.6.16) with $\rho = 2d_0$ (depending on the parameter d_0 !).

Assuming from (10.6.10) and (10.6.12) that

$$\widehat{d}_{lw} - d_0 \approx -\frac{1}{4m} \sum_{j=1}^m (z_j - \bar{z}) \left(\frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} - 1 \right),$$

where $z_j = -2 \log \lambda_j = -2 \log \frac{2\pi j}{N}$, and supposing that the difference between $f_{\eta_0}(\lambda_j)$ and $\mathbb{E} \widetilde{I}_X(\lambda_j)$ is negligible, we have

$$\widehat{d}_{lw} - d_0 \approx -\frac{1}{4m} \sum_{j=1}^m (z_j - \bar{z}) \left(\frac{\widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} - \frac{\mathbb{E} \widetilde{I}_X(\lambda_j)}{c_0 \lambda_j^{-2d_0}} \right) - \frac{1}{4m} \sum_{j=1}^m (z_j - \bar{z}) \left(\frac{f_{\eta_0}(\lambda_j)}{c_0 \lambda_j^{-2d_0}} - 1 \right). \quad (10.6.17)$$

The first term in (10.6.17) has zero mean, and its variance behaves as $1/(4m)$ by (10.6.9). The second term in (10.6.17), on the other hand, is deterministic and is responsible for the bias of $\widehat{d}_{lw}$. According to (10.6.16), it behaves as

$$\begin{aligned} \frac{1}{4m} \sum_{j=1}^m (z_j - \bar{z}) \left(\frac{f_{\eta_0}(\lambda_j)}{c_0 \lambda_j^{-2d_0}} - 1 \right) &\approx \frac{1}{4m} \sum_{j=1}^m (z_j - \bar{z}) c_\rho \lambda_j^\rho \\ &= \left(\frac{m}{N} \right)^\rho \frac{(2\pi)^\rho c_\rho}{4} \frac{1}{m} \sum_{j=1}^m (z_j - \bar{z}) \left(\frac{j}{m} \right)^\rho \approx -\left(\frac{m}{N} \right)^\rho \frac{(2\pi)^\rho c_\rho}{4} \frac{2}{m} \sum_{j=1}^m \left(\log \left(\frac{j}{m} \right) + 1 \right) \left(\frac{j}{m} \right)^\rho \\ &\approx -\left(\frac{m}{N} \right)^\rho \frac{(2\pi)^\rho c_\rho}{2} \int_0^1 (\log x + 1) x^\rho dx = -\left(\frac{m}{N} \right)^\rho \frac{(2\pi)^\rho c_\rho}{2} \frac{\rho}{(\rho + 1)^2}, \end{aligned} \quad (10.6.18)$$

where we used Proposition 2.10.2 to replace $z_j - \bar{z}$ and integration by parts in order to compute the integral above. The mean squared error (MSE) of $\widehat{d}_{lw}$ is then approximately equal to

$$\mathbb{E}(\widehat{d}_{lw} - d_0)^2 \approx \frac{1}{4m} + \frac{\rho^2 c_\rho^2 (2\pi)^{2\rho}}{4(\rho + 1)^4} \left(\frac{m}{N} \right)^{2\rho}, \quad (10.6.19)$$

where the first term accounts for the asymptotic variance and the second term for (the square of) the asymptotic bias. The optimal bandwidth m that balances the variance and the bias is (by differentiating and setting to 0 to solve for m)

$$m_{opt} = \left(\frac{(\rho + 1)^4}{2\rho^3 c_\rho^2 (2\pi)^{2\rho}} \right)^{\frac{1}{2\rho+1}} N^{\frac{2\rho}{2\rho+1}}. \quad (10.6.20)$$

Note also that $m = m_{opt}$ is such that, up to a multiplicative constant, the asymptotic variance $1/(4m)$ is equal to the asymptotic bias squared, expressed through the last term in (10.6.19). Indeed, this fact follows since by differentiating (10.6.19), the powers of m in both terms are reduced by 1.

Note that m_{opt} is of little use when ρ is unknown. If one is willing to assume a smooth model corresponding to $\rho = 2$, then

$$m_{opt} = KN^{\frac{4}{5}}, \quad (10.6.21)$$

where K is a constant depending on an unknown c_ρ . The constant K or c_ρ can be estimated through an iterative procedure (Henry and Robinson [470], Henry [468]) or a least squares method (Delgado and Robinson [299], Henry [468]).

Note also that with (10.6.21) when $\rho = 2$ or with (10.6.20) for general ρ , the bias is of the order $1/\sqrt{m}$ (see the observation following (10.6.20)) and thus is not negligible in the limit when multiplied by $\sqrt{m}$ (cf. Theorem 10.6.1). When $\rho = 2$ and m is m_{opt} in (10.6.21), the limit result (10.6.9) becomes

$$\sqrt{m_{opt}}(\widehat{d}_{lw} - d_0) - \frac{4\pi^2 c_\rho K^2}{9} \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{4}\right), \quad (10.6.22)$$

where the bias $\frac{4\pi^2 c_\rho K^2}{9}$ is that in (10.6.18) with $\rho = 2$, $m = KN^{\frac{4}{5}}$ and the change of sign in view of (10.6.17) (see also Corollary 1 and Comment 1 in Andrews and Sun [34]). In practice, after estimating c_ρ , the bias can be removed by considering $\widehat{d}_{lw} - 4\pi^2 c_\rho K^2 / (9\sqrt{m_{opt}})$.

10.6.3 Bias Reduction and Rate Optimality

The bias $\frac{4\pi^2 c_\rho K^2}{9}$ in (10.6.22) is not only characteristic of the bandwidth (10.6.21). In fact, it creates a well-known challenge to estimating d through the local Whittle method. According to (10.6.19), the bias is approximately

$$\frac{\rho c_\rho (2\pi)^\rho}{2(\rho + 1)^2} \left(\frac{m}{N} \right)^\rho$$

and can be nonnegligible for a range of bandwidths m .

If one is willing to assume $\rho = 2$, then the bias can be removed after estimating c_ρ as discussed above. Another possibility is to use tapering where the periodogram is replaced by

$$\frac{1}{2\pi \sum_n h_n^2} \left| \sum_{n=1}^N X_n h_n e^{-in\lambda} \right|^2, \quad (10.6.23)$$

where h_n is the taper, e.g., the cosine bell taper

$$h_n = 0.5 \left(1 - \cos \frac{2\pi(n - 1/2)}{N} \right), \quad n = 1, 2, \dots, N.$$

Tapering is well-known to reduce bias in estimating the spectral density, and leads to smaller bias in the local Whittle estimation of d (Hurvich and Chen [506]). One downside is that the asymptotic variance gets slightly inflated.

Another interesting approach to bias reduction was suggested by Andrews and Sun [34]. Assume that the spectral density satisfies

$$f_{\eta_0}(\lambda) = c_0 \lambda^{-2d_0} \left(1 + \sum_{k=1}^q \frac{b_k \lambda^{2k}}{(2k)!} + o(\lambda^{2q}) \right), \quad (10.6.24)$$

as $\lambda \rightarrow 0$. This coincides with the specification (10.6.16) with $\rho = 2q$. When $q = 1$, we get $\rho = 2$. The choice $q \geq 2$ in (10.6.24) assumes that $f_{\eta_0}(\lambda) \lambda^{2d_0}$ is smoother. The bias-reduced local Whittle estimator is obtained by minimizing

$$Q(c, d, \theta_1, \dots, \theta_{q-1}) = \frac{1}{m} \sum_{j=1}^m \left(\log(c \lambda_j^{-2d} e^{\sum_{k=1}^{q-1} \theta_k \lambda_j^{2k}}) + \frac{\tilde{I}_X(\lambda_j)}{c \lambda_j^{-2d} e^{\sum_{k=1}^{q-1} \theta_k \lambda_j^{2k}}} \right)$$

(cf. (10.6.3)). The estimator is also called the *local polynomial Whittle estimator*, in analogy to local polynomial kernel smoothing (Fan and Gijbels [353]). As with local polynomial kernel smoothing, the basic idea is that the log of the spectral density is now modeled around 0 as

$$\log(c \lambda_j^{-2d} e^{\sum_{k=1}^{q-1} \theta_k \lambda_j^{2k}}) = \log c - 2d \log \lambda_j + \sum_{k=1}^{q-1} \theta_k \lambda_j^{2k}.$$

The bias in estimating d is reduced by capturing deviations from a straight line in d in the log-log plot through a higher-order polynomial.

Under (10.6.24), the bias of the local polynomial Whittle estimator is of the order $(m/N)^{2q}$, which decreases as a power of (m/N) as q increases. On the downside, however, the asymptotic variance of the local polynomial Whittle estimator increases as $q \rightarrow \infty$ (Andrews and Sun [34], Comment 1 on p. 582), making the choice between a reduced bias and a larger asymptotic variance quite delicate. Usually a large sample size is needed to benefit from the bias-reduced estimation.

Finally, we also note that the issue of bias is naturally related to the rate of convergence of the local Whittle estimators. With the optimal bandwidth (10.6.20), the optimal rate of convergence is

$$m_{opt}^{-1/2} \quad \text{or} \quad N^{-\frac{\rho}{1+2\rho}}. \quad (10.6.25)$$

For example, when $\rho = 2$, the rate is $N^{-2/5}$. When the bias is reduced, one can expect a better convergence rate. Indeed, by balancing the orders of the asymptotic variance $1/m$

and the squared bias $(m/N)^{4q}$ as indicated above,⁷ the optimal bandwidth is $m_{opt} = CN^{4q/(1+4q)}$. The rate of convergence $m_{opt}^{-1/2}$ is

$$N^{-\frac{2q}{1+4q}}.$$

Note that this rate approaches the parametric rate $N^{-1/2}$ as $q \rightarrow \infty$.

10.7 Broadband Whittle Approach

In modeling LRD time series, the spectral density can be viewed as

$$f_{\eta}(\lambda) = \lambda^{-2d} f^*(\lambda), \quad (10.7.1)$$

where $f^*(\lambda)$ approaches a constant c as $\lambda \rightarrow 0$ (and λ^{-2d} can be replaced by $|1 - e^{-i\lambda}|^{-2d}$). Here, λ^{-2d} accounts for long-range dependence and $f^*(\lambda)$ captures short-range dependence effects. As in Sections 10.1 and 10.2, we can assume a parametric form for $f^*(\lambda)$, e.g., that of $\text{ARMA}(p, q)$ family. The MLE of d then converges to the true value at the parametric rate $N^{-1/2}$. This is a *parametric* and *global* approach. Alternatively, as in Section 10.6, the behavior of $f^*(\lambda)$ can be specified around $\lambda = 0$ only. Assuming $f^*(\lambda)$ is smooth at $\lambda = 0$, d can then be estimated at the rate $N^{-2/5}$ as in (10.6.25) with $\rho = 2$. This is a *semi-parametric* or *local (narrow-band)* approach.

There is one other interesting approach, called *broadband*, where $f^*(\lambda)$ is estimated by using a truncated series expansion. More specifically, suppose that $\log f^*(\lambda)$ has a Fourier cosine expansion

$$f^*(\lambda) = \exp \left\{ \sum_{l=0}^{\infty} a_l \cos(l\lambda) \right\}, \quad (10.7.2)$$

which can also be written as⁸

$$f^*(\lambda) = \frac{\sigma^2}{2\pi} \exp \left\{ \sum_{l=1}^{\infty} a_l \cos(l\lambda) \right\} \quad (10.7.3)$$

(after setting $\exp\{a_0\} = \sigma^2/(2\pi)$). By truncating the series in (10.7.2)–(10.7.3) to a finite order p , replace the Whittle approximation of the log-likelihood in (10.2.5) by

$$\ell_{bw,p}(\eta) = \sum_{j=1}^N \log \left(2\pi f_{p,\eta} \left(\frac{2\pi j}{N} \right) \right) + \sum_{j=1}^N \frac{I_X \left(\frac{2\pi j}{N} \right)}{2\pi f_{p,\eta} \left(\frac{2\pi j}{N} \right)}, \quad (10.7.4)$$

where $\eta = (d, a_0, a_1, \dots, a_p)$ and

$$f_{p,\eta}(\lambda) = \lambda^{-2d} \exp \left\{ \sum_{l=0}^p a_l \cos(l\lambda) \right\}.$$

⁷ Again, minimize $C_1/m + C_2(m/N)^{4q}$ with respect to m by differentiating and setting to 0.

⁸ σ^2 in (10.7.3) plays the same role as in the relation (10.2.6) and the discussion following it.

The objective function $\ell_{bw,p}(\eta)$ approaches the discrete Whittle approximation of the log-likelihood in (10.2.5) as $p \rightarrow \infty$ since $f_{p,\eta}(\lambda)$ becomes $\lambda^{-2d} f^*(\lambda)$. The function $\ell_{bw,p}(\eta)$ is convex in its parameters (Exercise 10.8), and is amenable to numerical minimization. The zero vector is often suggested for the initial estimators of the parameters for the nonlinear optimization. Set

$$(\widehat{d}_{bw}, \widehat{a}_0, \dots, \widehat{a}_p) = \widehat{\eta}_{bw} = \underset{\eta}{\operatorname{argmin}} \ell_{bw,p}(\eta). \quad (10.7.5)$$

In the next result, due to Narukawa and Matsuda [740], we provide the asymptotics of the broadband Whittle estimator $\widehat{d}_{bw}$ defined above. The following assumptions will be made:⁹

- (C1) The Fourier series (10.7.2) of $\log f^*(\lambda)$ is such that $\sum_{l=1}^{\infty} l^{\varrho} |a_l| < \infty$, with $\varrho > 9/2$.
- (C2) The series $\{X_n\}_{n \in \mathbb{Z}}$ can be expressed through a linear representation $X_n = \mu + \sum_{j=0}^{\infty} \alpha_j \epsilon_{n-j}$, $\sum_{j=0}^{\infty} \alpha_j^2 < \infty$, where $\{\epsilon_n\}$ is a sequence of i.i.d. random variables with $\mathbb{E}\epsilon_n = 0$, $\mathbb{E}\epsilon_n^4 < \infty$.
- (C3) The true parameter value $\eta_0 = (d_0, a_{0,0}, a_{0,1}, \dots)$ lies in the interior of the parameter space $\{d \in [0, 1/2), a_0, a_1, \dots \in \mathbb{R}, |a_j| \leq Cj^{-\delta}\}$ where $0 < C < \infty$ and $\delta > 1$. Moreover, for any parameter values $(d_1, a_0^1, a_1^1, \dots)$ and $(d_2, a_0^2, a_1^2, \dots)$ from the parameter space with $d_1 \neq d_2$, the functions $\lambda^{-2d_1} \exp\{\sum_{l=1}^{\infty} a_l^1 \cos(l\lambda)\}$ and $\lambda^{-2d_2} \exp\{\sum_{l=1}^{\infty} a_l^2 \cos(l\lambda)\}$ are different on a set of λ s with positive Lebesgue measure.
- (C4) $(p \Rightarrow) p_N \rightarrow \infty$, $p_N/N \rightarrow 0$ is such that

$$\frac{p_N^{2\varrho}}{N} \rightarrow \infty, \quad \frac{p_N^9 (\log N)^4}{N} \rightarrow 0$$

with $\varrho > 9/2$.

Theorem 10.7.1 Under Assumptions (C1)–(C4) above,

$$\sqrt{\frac{N}{p_N}} (\widehat{d}_{bw} - d_0) \xrightarrow{d} \mathcal{N}(0, 1). \quad (10.7.6)$$

See Narukawa and Matsuda [740] for a proof. If $p_N = N^r$, the condition (C4) above is satisfied when $1/(2\varrho) < r < 1/9$. The convergence rate then is

$$N^{\frac{1}{2} - r},$$

which approaches the parametric rate $N^{1/2}$ as $r \rightarrow 0$, possible when $\varrho \rightarrow \infty$. Note also that requiring $\varrho \rightarrow \infty$ here is very different from supposing $q \rightarrow \infty$ in the local polynomial Whittle estimation in Section 10.6.3 (see (10.6.24) and the discussion thereafter). Indeed, under smoothness assumptions on f^* (Exercise 10.9), the coefficients a_j in (10.7.2) are expected to decay exponentially fast and hence satisfy the condition (C1) above for arbitrarily large ϱ .

⁹ Strictly speaking, in Narukawa and Matsuda [740], λ^{-2d} is replaced by $|1 - e^{-i\lambda}|^{-2d}$ in (10.7.1).

10.8 Exercises

The symbols * and ** next to some exercises are explained in Section 2.12.

Exercise 10.1 Corollary 2.4.4 provides an explicit formula for the autocovariance function (ACVF) of a FARIMA(0, d , 0) series. Provide an explicit formula for the ACVF of a FARIMA(0, d , 1) series. Also, suggest a numerical procedure for evaluating the ACVF of a FARIMA(1, d , 0) series.

Exercise 10.2 Suppose that $\{X_n\}_{n \in \mathbb{Z}}$ is a stationary time series with autocovariance function $\gamma_X(n)$ and constant mean μ_X . The Best Linear Unbiased Estimator (BLUE) of μ_X is defined as $\hat{\mu}_{blue} = \sum_{n=1}^N w_n Y_n$, $w_n \in \mathbb{R}$, such that $\text{Var}(\hat{\mu}_{blue})$ is minimal subject to the condition $\sum_{n=1}^N w_n = 1$ (ensuring $\mathbb{E}\hat{\mu}_{blue} = \mu_X$). Show that the weight vector $w = (w_1, \dots, w_N)'$ is given by

$$w = \frac{\Sigma_N \mathbf{1}}{\mathbf{1}' \Sigma_N \mathbf{1}},$$

where $\mathbf{1} = (1, \dots, 1)'$ and $\Sigma_N = (\gamma_X(j - k))_{j,k=1,\dots,N}$. Show also that $\text{Var}(\hat{\mu}_{blue}) = (\mathbf{1}' \Sigma_N \mathbf{1})^{-1}$. For moderate N and FARIMA(0, d , 0) series, compare numerically the ratio of $\text{Var}(\hat{\mu}_{blue})$ to $\text{Var}(\bar{X})$, where $\bar{X}$ is the sample mean.

Exercise 10.3 Consider the log-likelihood in (10.2.8) and suppose for simplicity that σ^2 is known. The objective is then to minimize

$$\int_{-\pi}^{\pi} \frac{I_X(\lambda)}{h_v(\lambda)} d\lambda$$

with respect to v . At the population level, that is, replacing $I_X(\lambda)$ by the spectral density, this is equivalent to minimizing

$$F(v) = \int_{-\pi}^{\pi} \frac{h_{v_0}(\lambda)}{h_v(\lambda)} d\lambda,$$

where v_0 refers to the true parameter value. Explain why the function $F(v)$ is expected to reach its minimum at $v = v_0$ under mild assumptions? *Hint:* Use Parseval's identity $(1/2\pi) \int_{-\pi}^{\pi} |\hat{r}(\lambda)|^2 d\lambda = \sum_{n=-\infty}^{\infty} r_n^2$, where $r_n = 1/2\pi \int_{-\pi}^{\pi} e^{in\lambda} \hat{r}(\lambda) d\lambda$ are the Fourier coefficients of $\hat{r}(\lambda)$.

Exercise 10.4 The entries $c_{rs}(v)$ of the limiting covariance matrix in (10.2.12) are sometimes written as

$$c_{rs}(v) = \int_{-\pi}^{\pi} h_v(\lambda) \frac{\partial^2 h_v^{-1}}{\partial \partial v_r \partial v_s}(\lambda) d\lambda.$$

Show that the two expressions are equivalent.

Exercise 10.5* In Example 10.2.4, the asymptotic variance was computed for the Whittle estimator of d in FARIMA(0, d , 0) model. What is the asymptotic variance of the Whittle estimator of d in FARIMA(p , d , q) models?

Exercise 10.6 The functions `arfima` and `fracdiff` in the respective R packages `arfima` and `fracdiff` (cf. Section 10.5) use different methods in implementing MLE. Fit a FARIMA(1, d , 1) model to `tmpyr` data from the `arfima` package using the function `fracdiff`, and compare the fitted model to that obtained by using the function `arfima` in Section 10.5.

Exercise 10.7 The R package `longmemo` contains the celebrated time series `NileMin` of yearly minimal water levels of the Nile river for the years 622 to 1281, measured at the Roda gauge near Cairo. Fit a FARIMA(p , d , q) model to this time series using the function `fracdiff` (cf. Section 10.5) and provide its forecasts for 20 steps ahead.

Exercise 10.8 Show that the function $\ell_{bw,p}(\eta)$ in (10.7.4) is convex in η .

Exercise 10.9 Suppose $f^*(\lambda)$ in (10.7.1) is the spectral density of ARMA(p , q) time series; that is,

$$f^*(\lambda) = \frac{\sigma^2}{2\pi} \frac{|(1 - s_1^{-1}e^{-i\lambda}) \dots (1 - s_q^{-1}e^{-i\lambda})|^2}{|(1 - r_1^{-1}e^{-i\lambda}) \dots (1 - r_p^{-1}e^{-i\lambda})|^2},$$

where the roots satisfy $|s_n|^{-1} < 1$, $n = 1, \dots, q$, and $|r_m|^{-1} < 1$, $m = 1, \dots, p$.¹⁰ Show that the Fourier coefficients of the function $\log f^*(\lambda)$ decay exponentially. *Hint:* Use the fact that $\log(1 - z) = -\sum_{l=1}^{\infty} z^l/l$ for complex $|z| < 1$.

10.9 Bibliographical Notes

Section 10.1: To implement MLE in practice, it is critical to be able to compute numerically the autocovariance function (ACVF) of the fitted model. As indicated in Example 10.1.1, for FARIMA(p , d , q) models, the calculation of ACVF was considered in Sowell [918], Chung [242], Bertelli and Caporin [136], Doornik and Ooms [326]. See also Luceño [649] for an earlier work, Sela and Hurvich [896], Pai and Ravishanker [786, 785], Tsay [961] for extensions to the multivariate case, and Chen, Hurvich, and Lu [224] for related work. Several of these references (Sowell [918], Doornik and Ooms [326]) also touch upon other practical implementation issues behind MLE. An issue especially relevant in the multivariate case is being able to parameterize the fitted models in such a way that they are stationary (and causal and invertible, if desired) for the considered and hence estimated parameter values; see Ansley and Kohn [38], Roy, McElroy, and Linton [872].

Section 10.2: The Whittle approximation of the likelihood in the spectral domain goes back to Whittle [1005, 1006], including the multivariate case. See also Calder and Davis [197] for a historical perspective. For univariate short-range dependent series, the asymptotics of exact ML and Whittle estimators are established in, for example, Hannan [451].

For long-range dependent series, Yajima [1022] was the first to study the asymptotics of exact ML and Whittle estimators for Gaussian FARIMA(0, d , 0) model (though the Gaussianity assumption was made for simplicity). These estimators were studied under quite

¹⁰ These conditions correspond to invertible and causal ARMA time series.

general assumptions for Gaussian series in Fox and Taqqu [368] and Dahlhaus [277, 278]. The case of linear time series was considered in Giraitis and Surgailis [399]. The latter work provides the basis for the proof of Theorem 10.2.1, as well as Theorem 3.8.1 in Giraitis et al. [406]. For related developments, see also Heyde and Gay [473] concerning the multiparameter case, Hosoya [487] concerning multivariate time series, Velasco and Robinson [988] for extensions to the non-stationary case, and Andrews and Lieberman [33], Lieberman and Phillips [627], Lieberman, Rousseau, and Zucker [630], Lieberman et al. [631] for finer studies of MLE. The Whittle estimation for Gaussian subordinated times series is considered in Giraitis and Taqqu [402].

The convergence of quadratic forms of long-range dependent series to a normal limit plays a central role in the proof of Theorem 10.2.1. The approach taken in the proof is due to Giraitis and Surgailis [399]. For other work related to quadratic forms of long-range dependent series, see Avram [58], Fox and Taqqu [369], Ginovian [390], Giraitis and Taqqu [401], Horváth and Shao [483], Ginovyan and Sahakyan [391], Wu and Shao [1014], Avram, Leonenko, and Sakhno [60], Bai, Ginovyan, and Taqqu [83, 84]. A different approach, based on Stein's method and the Malliavin calculus, is taken in Nourdin and Peccati [761].

The question about the finite-sample performance of the ML estimators is raised in Remark 10.2.5. Hauser [461] provides a simulation study comparing the performance of the exact maximum likelihood (EML) for the demeaned data, the associated modified profile likelihood (MPL) and the Whittle estimator with (WLT) and without (WL) tapered data. Quoting from the abstract of Hauser [461]: "The tapered version of the Whittle likelihood turns out to be a reliable estimator for ARMA and ARFIMA models. [...] The modified profile likelihood is [...] either equivalent to the EML or more favorable than the EML. For fractionally integrated models, particularly, it dominates clearly the EML. The WL has serious deficiencies for large ranges of parameters, and so cannot be recommended in general. The EML, on the other hand, should only be used with care for fractionally integrated models due to its potential large negative bias of the fractional integration parameter." For other studies of finite sample properties, see also Sowell [918], Cheung and Diebold [231], Taqqu and Teverovsky [949], Bisaglia and Guégan [160], Reisen, Abraham, and Lopes [843], Nielsen and Frederiksen [751], Boutahar, Marimoutou, and Nouria [179]. Some of these studies also include the comparison to "local" estimation methods, the local Whittle estimator in Section 10.6 being an example.

Section 10.3: For other studies on model selection for long-range dependent series, see Bhansali and Kokoszka [141], Bisaglia [158], Carbonez [202]. For related work on estimation in misspecified models, see Chen and Deo [221].

Section 10.4: For theoretical studies concerning prediction of long-range dependent series, see Bhansali and Kokoszka [142, 143], Godet [414, 415]. Empirical considerations can be found in Brodsky and Hurvich [187], Hurvich [503], Ellis and Wilson [349], Bhardwaj and Swanson [145]. Brodsky and Hurvich [187], Basak, Chan, and Palma [105] compare the prediction performance of competing LRD and ARMA(1,1) models in both theory and applications. A critical look at the forecasting of FARIMA models can be found in Ellis and Wilson [349]; see also references therein.

For the behavior of the prediction coefficients, see also Inoue and Kasahara [521], Dębowski [292], and related interesting review papers by Bingham [155, 156].

Section 10.5: Several other R packages are available to analyze long-range dependent series. The package `longmemo` [895] implements some methods discussed by Beran [121]. The package `afmttools` [259] is another package for working with FARIMA models. The package `FGN` [975] provides synthesis and analysis tools around the fractional Gaussian noise model. See also the package `fArma` [1016] which includes a number of simulation and “local” estimation methods.

Time series modeling based on FARIMA models is also available through the commercial SAS/IML software [516].

Two temperature datasets were considered in Section 10.5 for illustration purposes. There are many other applications where long-range dependent time series are used. We shall not provide here a comprehensive review of all the many applications. We shall indicate next a few works of historical and other interests, where the focus is on the application at hand and model fitting in particular, rather than on methodological issues.

For applications and model fitting in hydrology, see, for example, Hosking [485], Noakes, Hipel, McLeod, Jimenez, and Yakowitz [754], Montanari, Rosso, and Taquq [730]. For applications in internet traffic modeling, see, for example, Scherrer et al. [887] where the reference is chosen for a more explicit modeling of the traffic, as opposed to the many empirical studies providing evidence of long-range dependence. For applications in economics and finance, see, for example, Diebold and Rudebusch [311], Cheung [230], Harvey [457]. Another often cited and early application to wind power can be found in Haslett and Raftery [458].

Section 10.6: The local Whittle estimator was suggested in Künsch [582], and justified theoretically in the seminal work by Robinson [853]. (As indicated in Section 10.6.1, the estimator was considered under slightly weaker assumptions in Andrews and Sun [34].) Extensions to nonstationary long-range dependent series (associated with the value $d \geq 1/2$) are considered in Velasco [984], Shimotsu and Phillips [907, 908], Abadir, Distaso, and Giraitis [2]. When the Whittle estimator is viewed in the M -estimation framework, see Robinson and Henry [856].

In addition to Delgado and Robinson [299], Henry and Robinson [470], Henry [468] mentioned in Section 10.6.2, see also Hassler and Olivares [459] for a recent empirical study of bandwidth selection.

Bias reduction and rate optimality is discussed in a nice review paper by Henry [469]. Giraitis, Robinson, and Samarov [403, 404] is another influential but somewhat technical study on rate optimality (see also Hidalgo and Yajima [474]). For another work on tapering, see Chen [219]. The work of Andrews and Sun [34] is continued in Guggenberger and Sun [439]. See also Giraitis and Robinson [395], Lieberman and Phillips [627, 628] for Edgeworth expansions, Soulier [916] for best attainable rates in estimation, and Baillie et al. [90] for further discussion on bias.

Empirical studies and comparison of several available semi-parametric estimators, including the local Whittle estimator, can be found in the references given in the notes to Section 10.2.

Section 10.7: A broadband approach similar to that described in Section 10.7, where the exponential in (10.7.2) is replaced by the square of a Fourier transform itself, was taken

by Bhansali, Giraitis, and Kokoszka [144]. An application of the broadband local Whittle estimation to testing for long-range dependence can be found in Narukawa [739].

Other notes: Another local semi-parametric estimator, the GPH estimator, was given in Section 2.10.3. It should be noted that the two estimators, the GPH and the local Whittle, have been developed very much in parallel. For example, following the presentation for the Whittle estimator, the GPH estimator (possibly involving tapering) was studied by Hurvich and Ray [508], Velasco [985], Nouria, Boutahar, and Marimoutou [759]. The bias of the GPH estimator and the choice of the bandwidth were considered in Agiakloglou et al. [15], Hurvich and Beltrao [504], Hurvich and Deo [507], Hurvich et al. [510]. The local polynomial modification of the GPH estimator can be found in Guggenberger and Sun [439]. The broadband estimator associated with the GPH and based on fitting a fractional exponential (FEXP) model was studied in Moulines and Soulier [734], Hurvich [502], Hurvich and Brodsky [505], Hurvich, Moulines, and Soulier [511], Iouditsky, Moulines, and Soulier [523].

A number of other topics on estimation of long-range dependent time series were not included in Chapter 10. Some of these are discussed in connection to Chapters 2 and 9. Several others are listed below for interested readers and a few references are provided without aiming at being exhaustive.

Estimation of the so-called impulse response coefficients in LRD series is considered in Baillie and Kapetanios [88], Baillie, Kapetanios, and Papailias [91].

FARIMA modeling with stable innovations is studied in Kokoszka and Taqqu [571, 573], Burnecki and Sikora [194]. Nonlinear long-range dependent time series and their estimation, of particular interest in modeling stochastic volatility, are considered in Deo, Hsieh, Hurvich, and Soulier [302], Shao and Wu [902]. For seasonal long-range dependent models, see Arteche and Robinson [51], Oppenheim, Ould Haye, and Viano [781], Palma and Chan [790], Reisen, Rodrigues, and Palma [844], Philippe, Surgailis, and Viano [808]. State space modeling of long-range dependent series can be found in Chan and Palma [215].

It is known that long-range dependence is confused easily with non-stationary-like behavior, such as shifting means. Distinguishing the two phenomena has attracted a lot of attention. See, for example, Berkes et al. [130], Bisaglia and Gerolimetto [159], Iacone [513], Qu [837], Baek and Pipiras [69], McCloskey and Perron [701], Dehling, Roach, and Taqqu [297]. (See also the notes to Section 3.6 above.) Prediction performance of competing models, LRD and series with breaks, is investigated in Gabriel and Martins [378]. An approach to modeling trends, long-range dependence and nonstationarity, through SEMIFAR models, is considered in Beran and Feng [123]. See also Roueff and von Sachs [868], Sen, Preuss, and Dette [898].

Bayesian methods for long-range dependent time series are considered in Pai and Ravishanker [787], Liseo, Marinucci, and Petrella [638], Holan, McElroy, and Chakraborty [480], Graves, Gramacy, Franzke, and Watkins [425]. Robust estimation is discussed in Lévy-Leduc, Boistard, Moulines, Taqqu, and Reisen [619]. Parametric and nonparametric regressions with long-range dependent errors are considered in Sections 11 and 12 of Giraitis et al. [406], Section 7 of Beran et al. [127].

Bootstrap and other resampling schemes are used in Hall, Jing, and Lahiri [447], Bisaglia, Bordinon, and Cecchinato [161], Conti, De Giovanni, Stoev, and Taqqu [258], Kim

and Nordman [556, 557], Zhang, Ho, Wendler, and Wu [1028], Grose, Martin, and Poskitt [434], Poskitt [829], Poskitt, Grose, and Martin [830], Shao [901]. See also Section 10 in Beran et al. [127]. Subsampling, also called “block sampling,” is a form of bootstrap. When used together with self-normalization, it provides an effective method for approximating an asymptotic distribution function without having to know potential nuisance parameters, for example an unspecified Hurst index H when dealing with long-range dependent processes. For details, see Betken and Wendler [140], Bai, Taqqu, and Zhang [85], Bai and Taqqu [81].

Testing for long-range dependence in the time domain is considered in Hassler, Rodrigues, and Rubia [460]. Studies of the asymptotic distributions of the sample mean, autocovariances and autocorrelations of LRD series go back to Hosking [486]. See also Lévy-Leduc, Boistard, Moulines, Taqqu, and Reisen [620], Wu, Huang, and Zheng [1015]. Estimation of inverse autocovariances matrices is considered in Ing, Chiou, and Guo [517].

Finally, empirical graphical methods for estimating the long-range dependence parameter can be found in Taqqu, Teverovsky, and Willinger [951] and Taqqu and Teverovsky [948]. A wavelet Whittle estimator of the long-range dependence parameter is described in Moulines et al. [737], with the extension to the multivariate setting in Achard and Gannaz [13]. A comparison between Fourier and wavelet techniques can be found in Faÿ, Moulines, Roueff, and Taqqu [358].